Tail Certificates and Achievement-Set Geometry

Lean-checked exact reductions for Erdős #249 and the half-value branch of Erdős #257

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Abstract

For the totient constant $S = \sum_{n\geq 1} \varphi(n)/2^n$ of Erdős #249 and the Mersenne support sums $\sum_{n\in A}(2^n-1)^{-1}$ of Erdős #257, rationality would impose rigid discrete structure: eventually periodic binary tails for S, and an integral carry orbit with Boolean Möbius quotient stream for a support sum. We formalise both forms of rigidity in Lean and place #249 and the distinguished half-value counterexample route to #257 in exact finite-to-cofinal form: each open endpoint is proved equivalent to a cofinal supply of checkable finite events. The universal #257 assertion itself is not reduced to this chain. One certified Farey window excludes every rational representation $S = p/q$ with $q \leq Q_0 \approx 7.96 \times 10^{34}$; hence a rational representation must have $q > Q_0$. Irrationality of S is also equivalent to non-integrality of every positive binary tail difference, to complete finite certificates at fixed parameters, and to a cofinal lcm-diagonal certificate supply. Certificates are verified at the 28 scales up to $t = 64$ at which the lcm diagonal changes (including the initial scale); this is finite evidence only, and the cofinal supply is open. A CRT–Dirichlet construction also gives exact dyadic totient-kernel rank 2^e+1 at every level, so the full kernel span is infinite-dimensional. Rationality would supply a positive-multiplier tempered integral carry orbit whose canonical level-e carry-section span has dimension at least 2^e-1 for every e; no upper bound is known, so this does not prove irrationality.

For Erdős #257, we formalise Erdős's classical theorem for full support in every base and several structured-support cases. At base 2 we prove exact achievement-set geometry and necessary carry and gap constraints on hypothetical rational infinite supports; in particular, the divisor stream of a rational support admits only sublogarithmic zero windows, uniformly in the support. Membership of 1/2, which would refute the universal statement, is exactly equivalent to infinitely many greedy skips. Nonmembership is exactly equivalent to an eventually right integer seam, a finite fatal gap, and a final greedy skip.

The current final-skip route excludes a final upper transition and the middle transition carry -3. One global tail-dominance hypothesis remains: exclude the middle carries -2 and -1, and prove the future-tail inequality at the remaining nonnegative-carry middle transitions. An unbounded certificate supply for #249 and cofinal terminal zeros for #257 are not proved. The universal #257 statement and half-value membership remain open. Neither problem is settled.

1 Introduction

The two questions are whether $S = \sum_{n\geq 1} \varphi(n)/2^n$ is irrational (#249), and whether $\sum_{n\in A}(2^n-1)^{-1}$ is irrational for every infinite $A \subseteq \mathbb{N}$ (#257). The formulations occur in the Erdős–Graham collection and Erdős's later survey [3, 4]; the numbering follows the maintained catalogue [21].

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{N}_{>0} = \{1, 2, 3, \ldots\}$. Supports for #257 are subsets of $\mathbb{N}_{>0}$; allowing the exponent 0 would introduce the undefined denominator $2^0 - 1$.

For #257, an infinite support summing to 1/2 would refute the universal claim. Neither endpoint is proved. In both open spines, tail and carry identities separate exact finite arithmetic from one analytic or combinatorial cofinality statement.

Overview for a first reading

The totient $\varphi(n)$ is Euler's totient. A *support* selects terms; its *achievement set* contains the resulting sums. *Cofinally* means beyond every bound. A *spine* is a displayed chain of equivalences or implications leading to one open quantified statement.

The totient constant also has the elementary probabilistic form

$$S = \frac{1}{2} + \Pr(\gcd(X,Y)=1),$$

where X and Y are independent fair-coin waiting times. The #257 question asks whether every infinite choice of the fractions $1/(2^n-1)$ has irrational sum. The two exact finite-to-cofinal spines are

$$\begin{aligned} \text{cofinally many lcm-diagonal certificates} \quad &\Longleftrightarrow \quad S \notin \mathbb{Q}, \\ \text{cofinally many greedy terminal zeros} \quad &\Longleftrightarrow \quad \frac{1}{2} \in \mathcal{A} \\ &\Longrightarrow \quad \text{a counterexample to universal \#257.} \end{aligned}$$

For #249, a denominator exclusion and 28 diagonal certificates are proved, but no unbounded certificate supply is known. For the half-value branch of #257, finite-support exclusion and the exact greedy classification are proved, but no cofinal supply of terminal zeros is known. This shared quantifier pattern is the reason for treating the problems together; it is an architectural comparison, not a common theorem connecting their arithmetic mechanisms.

Question or result	Status	What is proved
Irrationality in #249	Open	No proof of $S \notin \mathbb{Q}$ is claimed.
Denominator exclusion	Proved	If $S = p/q$, then $q > Q_0$.
Tail and lcm-diagonal forms	Exact equivalences	Either certificate supply is equivalent to $S \notin \mathbb{Q}$; the supply itself is open.
Diagonal computations through $t = 64$	Finite evidence	Lean checks the 28 lcm-height changes in that range; this is not a cofinal theorem.
Universal assertion in #257	Open	No proof covers every infinite support.
Full and structured supports	Proved cases	Classical full-support irrationality and several restricted support families are formalised.
Rational-support carry laws	Necessary conditions	Any hypothetical rational support must satisfy the stated orbit, gap, mass, and unbounded-state constraints.
Membership of 1/2	Open	Membership would give an infinite rational support and refute universal #257.
Greedy and final-skip forms of 1/2	Exact equivalences	They identify the missing cofinal event but do not prove that it occurs.
Rank-26 final-middle sieve	Conditional reduction	The orbit restrictions and the exact count of 412 surviving phases are proved; the survivors are not excluded.

What "Lean-checked" means here. A linked theorem statement has a proof term accepted by Lean from the pinned source and dependencies. An exact equivalence remains an equivalence even when both sides are open; a finite calculation remains finite; and a theorem with a rationality hypothesis does not prove that a rational support exists. The manuscript, its link checker, and its generated indexes help a reader find the proof but are not themselves proof authority. The release section states the additional source-pin and trust checks required for a reproducible copy.

Principal results and contribution. Four theorem families carry the paper:

(i) the denominator exclusion in Theorem 5.1 and the infinite-dimensional dyadic totient-kernel theorem in Theorem C.15;

(ii) the fixed, pointwise, cofinal, and lcm-diagonal certificate equivalences in Proposition 5.4 and Theorems 5.5–5.7;

(iii) the infinite state, zero-window, and denominator-mass constraints in Corollaries 4.5–4.8;

(iv) the half-membership and final-skip classifications in Theorems 4.11 and 4.14, with the residual obligations in Proposition 4.15 and Corollary 4.16.

The table above separates proved results, exact reformulations, conditional implications, finite evidence, and open endpoints. In particular, the 28 diagonal rows remain finite evidence, and both endpoints remain open.

These results are proved here without a priority claim. The main article is the expository entry point; appendices carry auxiliary and audit material. Neither evidence nor an exact reduction supplies the missing unbounded witness family. The exact half-value counterexample spine uses that a finite normalised support cannot have value $1/2$; no unbounded supply is proved.

Reading map. Section 2 gives the exact spines, Section 3 their common coordinates, and Sections 4 and 5 the results, verification, and frontiers. Appendices A and B contain the local half-value and carry arguments; later appendices treat the totient certificate and Lambert structure. Linked phrases open exact Lean proofs. The former one-sided #249 companion is retired; its checked transport and first-harmonic claims remain available through the registry without changing the problem's open status.

Keywords. irrationality; Erdős–Borwein constant; Euler totient; Lambert series; achievement sets; binary carry systems; Lean 4. **MSC 2020.** 11J72 (primary); 11A25, 68V20 (secondary).

2 The two exact spines

2.1 Erdős #249: an exact certificate normal form

Write

$$R_N = \sum_{m \geq 1} \frac{\varphi(N+m)}{2^m}.$$

Each R_N is $2^N S$ minus an integer. Rationality of S would force one binary period h whose tail differences $R_{N+h} - R_N$ are eventually all integral; the certificates below refute exactly such candidate periods (Section 5.3).

Exact reduction. The formal source proves the exact chain

$$\begin{aligned} S \notin \mathbb{Q} \quad &\Longleftrightarrow \quad R_{N+h} - R_N \notin \mathbb{Z} \quad \text{for every } h > 0 \text{ and every } N \\ &\Longleftrightarrow \quad \text{for every } h > 0, N \text{ there is an } L \text{ with } \mathsf{Sep}(h, N, L). \end{aligned}$$

Here $\mathsf{Sep}(h, N, L)$ is a finite residue computation with a rigorous tail allowance. The second equivalence is completeness of the certificate: at fixed h, N, some finite depth succeeds exactly when the tail difference is non-integral. Conversely, integrality of one positive-shift difference forces S to be rational. The three statements are formalised by the tail-difference equivalence, the pointwise certificate equivalence, and the fixed-parameter certificate-completeness theorem.

Unconditional finite baseline. Independently of the unbounded certificate supply, if $S = p/q$, then

$$q > Q_0 = 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053.$$

This is the denominator exclusion.

Missing theorem. An unbounded certificate supply is the exact open statement, equivalently an unbounded supply of non-integral tail differences. Theorem 5.7 compresses this to the equivalent one-parameter lcm-diagonal supply. Finite diagonal instances are verified at the 28 changing-lcm scales through $t = 64$, but no unbounded supply is proved. Neither the finite denominator exclusion nor the verified diagonal instances discharge this quantifier.

2.2 Erdős #257: the exact half-value counterexample spine

Let

$$\mathcal{A} = \left\{ \sum_{n \in A} \frac{1}{2^n - 1} : A \subseteq \mathbb{N}_{>0} \right\}.$$

Exact reduction and finite exclusion. The weights are strictly superincreasing, so the normalised support of a value is unique. A finite normalised support cannot have value $1/2$. For $1/2$, the following are equivalent:

$$\begin{aligned} \frac{1}{2} \in \mathcal{A} \quad &\iff \quad \text{the set of exponents omitted by the greedy expansion is infinite} \\ &\iff \quad \text{a successor terminal bit equal to zero occurs beyond every bound.} \end{aligned}$$

Concretely, the greedy expansion of $1/2$ skips rank 1, takes ranks $2, 3$, skips $4, 5$, and takes $6, 7$; membership asks whether such skips recur beyond every bound.

Counterexample consequence. Membership would produce an infinite support of rational sum and refute the universal statement of Erdős #257. The required counterexample condition is therefore exact: cofinally many terminal zeros, equivalently no last greedy skip. The reduction and finite exclusion are formalised by the infinite-skip equivalence, unbounded-terminal equivalence, no-last-skip equivalence, and the finite-support exclusion.

Missing theorem. The missing input is a proof that terminal zeros occur cofinally. The strongest current last-skip route is conditional. After a putative last skip, the upper transition and the middle coordinate -3 are impossible. The two final-skip obligations are to exclude the actual middle coordinates $-2, -1$ and establish the exact future-tail inequality at every other non--3 middle transition. Neither is proved for the greedy orbit, so this route does not yet produce cofinal terminal zeros. The conditional residual is the remaining-tail implication. Its local transition exclusions are stated in Proposition 4.15, and its sufficient global hypothesis is Corollary 4.16. Appendix A.4 supplies the exact final-skip coordinates consumed by those two statements. Appendices A.2–A.3 record alternative conditional half-value criteria, while Appendix A.5 supplies bounded evidence and closes invalid shortcuts; none discharges the residual above.

Prior-art boundary. Full support is Erdős's, and Borwein proved a broader theorem containing the same specialisation [1, 5]. Pairwise-coprime and periodic-support results are represented by Erdős and Luca–Tachiya [2, 13]. Strict-tail and carry precedents are Kovač–Tao and Wang–Grau Ribas [14, 15]; Crandall and Campbell address adjacent binary-expansion questions [16, 17]. The maintained record contains no exact antecedent for the denominator bound, certificate equivalence, or last-skip reduction; that is not itself a novelty claim.

3 The Mersenne–Lambert ladder

Define the Lambert-type transform of an arithmetic function f by

$$L(f) = \sum_{n\geq 1} \frac{f(n)}{2^n - 1} = \sum_{m\geq 1} \frac{(f * \mathbf{1})(m)}{2^m}, \qquad (f * \mathbf{1})(m) = \sum_{d|m} f(d),$$

the second equality being the standard Lambert rearrangement (one Dirichlet convolution by the constant $\mathbf{1} = \zeta$); Apostol supplies the classical arithmetic-function and Dirichlet-convolution background [6]; Merca introduced a modern Lambert-series factorisation theorem [7], and Merca–Schmidt give a synthesis/unification treatment [8]. Three inputs form the convolution chain

$$\alpha \xrightarrow{*\mathbf{1}} \varphi \xrightarrow{*\mathbf{1}} \mathrm{Id}, \qquad \alpha = \varphi * \mu,$$

whereas μ and $\mathbf{1}$ are comparison inputs. The corresponding values have the following statuses.

input f	value	mathematical status	treatment here
Convolution chain			
$\alpha = \varphi * \mu$	$L(\alpha) = S$	open (#249)	identities and bounds formalised
φ	$L(\varphi) = 2$	rational	formalised
Id	$L(\mathrm{Id}) = \sum \sigma(m)/2^m$	transcendental	cited [11]
Comparison inputs			
μ	$L(\mu) = 1/2$	rational	formalised
$\mathbf{1}$	$L(\mathbf{1}) = E$	irrational	formalised

The weight α is the primitive-conductor count: $\alpha(p) = p - 2$ on primes and $\alpha \geq 0$. The chain records $\alpha * \mathbf{1} = \varphi$ and $\varphi * \mathbf{1} = \mathrm{Id}$. Its associated values are open at $L(\alpha) = S$, rational at $L(\varphi) = 2$, and cited transcendental at $L(\mathrm{Id}) = \sum_m \sigma(m)/2^m$. The rows $L(\mu) = 1/2$ and $L(\mathbf{1}) = E$ are not further steps in this chain. In particular, S and E have the same Mersenne-denominator form, with weights α and $\mathbf{1}$, but passing between those weights is not a convolution step.

Proposition 3.1 (Möbius rational rung)**.**

$$\sum_{d\geq 1} \frac{\mu(d)}{2^d - 1} = \frac{1}{2}.$$

Formalised as the Möbius rung.

Proposition 3.2 (totient rational rung)**.**

$$\sum_{d\geq 1} \frac{\varphi(d)}{2^d - 1} = 2.$$

Formalised as the totient rung.

Proposition 3.3 (positive lift)**.** *With the primitive-conductor weight $\alpha = \varphi * \mu$,*

$$\sum_{d\geq 1} \frac{\alpha(d)}{2^d - 1} = \sum_{n\geq 1} \frac{\varphi(n)}{2^n} = S.$$

The first equality is formalised as the positive-lift identity. The second is the definition of S. The proposition places S in the same $\sum(\cdot)/(2^d - 1)$ family as E, with the nonnegative weight α.

Probabilistic reading. If X, Y are independent fair-coin waiting times, $\Pr(X = n) = 2^{-n}$, then

$$S = \tfrac{1}{2} + \Pr(\gcd(X, Y) = 1).$$

Thus Erdős #249 also asks whether the probability that two independent fair-coin waiting times are coprime is irrational. The exact identity is Proposition 5.2; its Möbius-square form and the periodic and finite-algebraic variants of the ladder are collected in Appendices D.1–D.2.

Section consequence and residual. The exact contribution of this section is the coordinate change $L(\alpha) = S$, together with the two separate rational-rung propositions above. The convolution chain relates S to rational and transcendental neighbouring values, while the comparison inputs place $1/2$ and E in the same transform. No status transfers from one input to another. For #249 the remaining issue is still the unbounded certificate supply isolated in Section 5; for #257 the ladder supplies context rather than either the universal theorem or a rational counterexample.

4 Erdős–Borwein-type irrationality (the #257 direction)

There are two logically distinct directions. The universal direction asks for irrationality for every infinite support; the counterexample direction asks for one infinite support with rational value, and the target $1/2$ gives an exact formulation of that problem. This section first states the classical and structured irrationality theorems, then gives the base-2 achievement-set geometry and the constraints forced by a hypothetical rational value. It ends with the exact classification of the target $1/2$ and the exact residual condition left by the current final-skip argument. The final-skip theorems consume Appendix A.4; Appendices A.2–A.3 contain alternative conditional criteria; and Appendix A.5 contains finite evidence, coordinate changes, and closed branches. These are auxiliary implications, not additional conclusions of this section.

4.1 Full support: the Erdős–Borwein constant, every base

Theorem 4.1 (full-support irrationality). *For every integer $b \geq 2$, the series $\sum_{n\geq 1} \frac{1}{b^n - 1}$ is irrational. In particular the Erdős–Borwein constant $E = \sum_{n\geq 1} 1/(2^n - 1)$ is irrational.*

The all-base theorem and its base-2 corollary are formalised. This is Erdős's 1948 theorem [1]. In the ladder of Section 3 it is the rung $L(\mathbf{1})$, and it is the fixed machine-checked comparison point in the same Mersenne–Lambert family as the open S; as Section 3 notes, it is not a further convolution step from S.

The proof begins with the Lambert identity

$$\sum_{n\geq 1} \frac{1}{b^n - 1} = \sum_{m\geq 1} \frac{\tau(m)}{b^m},$$

where $\tau(m)$ is the number of positive divisors of m; see the Lambert identity. If the right-hand side were rational with reduced denominator q, then a non-integral integer multiple could not lie at distance less than $1/q$ from an integer. It is therefore enough, for every q, to construct N such that $b^N \sum_m \tau(m) b^{-m}$ is non-integral but lies within $1/q$ of an integer. This is the formal near-integer criterion.

The construction of N is divided into three finite parts. First, a bounded Bertrand argument supplies disjoint prime blocks, and the Chinese remainder theorem chooses an arithmetic progression on which

$$b^r \mid \tau(N + r) \qquad (1 \leq r \leq K).$$

These divisibilities make the first K terms of the shifted tail integral. The multiplicativity step that converts prescribed prime valuations into the displayed divisibility is the prime-block divisibility lemma. Second, divisor pairing bounds the average of the weighted middle window

$$\sum_{K<r\leq L} \tau(N+r)b^{L-r}$$

along that progression. A pigeonhole choice then selects one translate with a small middle contribution; see the weighted-middle selection lemma. Third, the elementary estimate $\tau(n) \leq n$ controls the infinite tail after L. The parameters are chosen so that the middle and far-tail bounds together are smaller than $1/q$; the shifted tail is strictly positive because every divisor count is positive, so the dilated sum is not an integer.

Thus the analytic series is consumed only at the Lambert identity and the geometric tail estimate. The remaining proof is a finite CRT construction, divisor counting, and explicit parameter arithmetic. The final certificate existence theorem holds for every $b \geq 2$ and every precision q; see the uniform certificate theorem. Its composition with the near-integer criterion is the theorem cited above. This is a formalisation of the classical full-support theorem, not a new irrationality result.

4.2 Named infinite-support cases

Beyond full support, the development proves irrationality for a family of infinite supports A, at every base $b \geq 2$. Each is a formalised case of the #257 statement family. None is the universal statement, and none is claimed as new mathematics.

Theorem 4.2 (support-class family). *For every integer $b \geq 2$, the series $\sum_{n\in A} 1/(b^n-1)$ is irrational for each of the following infinite supports A:*

(a) *factorial support $A = \{n! : n \geq 1\}$, giving $\sum 1/(b^{n!}-1)$;*

(b) *power-of-two support $A = \{2^n : n \geq 0\}$, giving $\sum 1/(b^{2^n}-1)$;*

(c) *multiples $A = d\mathbb{N}$ with $d \geq 1$;*

(d) *pairwise-coprime supports with summable reciprocals (Erdős's condition [2]);*

(e) *eventually-periodic supports, residue classes, and the odd numbers.*

These sort into three groups. The classical case is (d): infinite pairwise-coprime A with $\sum_{a\in A} 1/a < \infty$, formalised from Erdős's condition. The lcm-gap named cases are (a) and (b), whose support gaps outgrow the running lcm; the structured families are (c) and (e).[1] Formalised case by case by the factorial-support theorem, power-of-two-support theorem, multiple-support theorem, pairwise-coprime-support theorem, and eventually-periodic-support theorem. The residue-class and odd supports in part (e) are direct specialisations of the last theorem; the first two theorems already hold for every base $b \geq 2$.

The prime support is now known unconditionally: Tao and Teräväinen proved that

$$\sum_p \frac{1}{2^p-1} = \sum_{n\geq 1} \frac{\omega(n)}{2^n}$$

is irrational [18]. Their proof uses quantitative two-point correlations of multiplicative functions and is not formalised here. This settles an important special case, while the universal support statement remains open.

[1] Luca–Tachiya prove the broader eventual-periodic rational-coefficient theorem; its indicator specialisation directly contains case (e) [13]. Their Chowla–Erdős-style large-modulus proof is distinct from the periodic-divisor certificate route formalised here, and the full mixed-sign coefficient theorem is not claimed.

The factorial and power-of-two cases are worth a word because they are not reachable by a Liouville shortcut: the power-of-two gap $2^k - \mathrm{lcm}$ stays within a bounded ratio of the lcm, so the terms do not decay fast enough for a transcendence-measure argument, and a uniform criterion is needed. The mechanism is *period noncollapse*: a prime-valuation-deficit witness forces the multiplicative order of the base modulo the relevant modulus not to collapse, which rules out the long repeated digit blocks a rational value would require.

Transition to the counterexample direction. The results above enlarge the family of supports for which irrationality is known, but they do not supply the universal quantifier. We now fix the base at 2 and change coordinates: all support sums form one achievement set. The counterexample problem thereby becomes a point-membership question, with generic constraints on its rational points and a separate exact reduction at the target $1/2$.

4.3 The base-2 achievement set

For $n \geq 1$, put

$$w_n = \frac{1}{2^n - 1}.$$

For $n \geq 0$, put

$$T_n = \sum_{m>n} w_m, \qquad \mathcal{A} = \left\{ \sum_{n\in A} w_n : A \subseteq \mathbb{N},\ 0 \notin A \right\}.$$

For $x \geq 0$, define the greedy residuals by $r_0(x) = x$ and

$$\varepsilon_n(x) = \begin{cases} 1, & w_n \leq r_{n-1}(x), \\ 0, & w_n > r_{n-1}(x), \end{cases} \qquad r_n(x) = r_{n-1}(x) - \varepsilon_n(x) w_n.$$

When $\varepsilon_n(x) = 1$ we say the expansion *takes* rank n; otherwise it *skips* rank n.

Theorem 4.3 (strict-tail greedy coding)**.** *For every $n \geq 1$,*

$$T_n < w_n.$$

Hence every $x \in \mathcal{A}$ has a unique normalised support. Moreover, a real number x lies in $\mathcal{A}$ if and only if $x \geq 0$ and $r_n(x) \leq T_n$ for every $n \geq 0$. In that case $x = \sum_{n\geq 1} \varepsilon_n(x) w_n$.

The strict-tail inequality and greedy criterion are the strict-tail theorem and greedy membership theorem. Uniqueness is the support-coding theorem. Kovač–Tao provide the strict-tail Cantor-set context [14].

Separate geometric and finite-arithmetic consequences. Proposition B.12 proves that $\mathcal{A}$ is compact, perfect, totally disconnected, nowhere dense, and of Lebesgue measure 1. Proposition B.13 proves soundness of exact rational evaluation and gives finite nonmembership certificates. These results use the same greedy coordinates, but neither is a clause of the membership equivalence above. Their detailed statements and one-sided boundary are in Appendix B.5. No novelty claim is made for the strict-tail geometry.

Consequence for the half-value branch. Theorem 4.3 makes the greedy support canonical; Proposition B.12 supplies the compactness needed to pass from cofinal finite approximations to a point of $\mathcal{A}$; and Proposition B.13 supplies one-sided finite refuters. The measure-one statement is global and does not decide whether the particular point $1/2$ belongs to $\mathcal{A}$. Before specialising to that point, we record the carry constraints forced by any rational $X_A \in \mathcal{A}$; these are necessary conditions in the universal direction, not yet a contradiction.

4.4 Rigidity of a hypothetical rational support

For $A \subseteq \mathbb{N}_{>0}$, define

$$f_A(m) = \#\{a \in A : a \mid m\}, \qquad X_A = \sum_{a\in A} \frac{1}{2^a - 1} = \sum_{m\geq 1} \frac{f_A(m)}{2^m}, \qquad T_f(N) = \sum_{j\geq 1} \frac{f(N+j)}{2^j}.$$

Theorem 4.4 (rational-support carry recurrence)**.** *Suppose that $A \neq \varnothing$ and*

$$X_A = \frac{p}{2^c v}, \qquad p \in \mathbb{Z}, \quad c \in \mathbb{N}, \quad v \geq 1 \text{ odd.}$$

Then $u_N = vT_{f_A}(c+N)$ is a positive integer for every N, and

$$u_{N+1} + vf_A(c+N+1) = 2u_N. \tag{4.1}$$

This is formalised by the shifted natural-state theorem. The recurrence is binary long division: since $2T_f(M) = f(M+1) + T_f(M+1)$, doubling the scaled tail surrenders the next coefficient, and rationality is what makes the states u_N integers. Its tail-orbit normal form is Theorem B.1.

Corollary 4.5 (unbounded-state consequence)**.** *Under the hypotheses of Theorem 4.4, if A is infinite, then (u_N) is unbounded.*

This is formalised by the fraction-facing unbounded-state theorem; see also Theorem B.10.

Corollary 4.6 (sublogarithmic coverage consequence)**.** *Under the hypotheses of Theorem 4.4, for every $\varepsilon > 0$ there is $B = B(\varepsilon, c, v)$ such that every interval $\{c+N+1, \dots, c+N+\ell\}$ on which f_A vanishes satisfies*

$$\ell \leq \varepsilon \log_2(N+1) + B;$$

This is formalised by the sublogarithmic zero-window theorem; see also Theorem B.6.

Corollary 4.7 (odd-denominator reciprocal-mass consequence)**.** *Under the hypotheses of Theorem 4.4, if $\rho(A) = \sum_{a\in A} a^{-1} < \infty$, $v > 1$, and $h = \operatorname{ord}_v(2)$, then*

$$\rho(A) = \frac{w}{h} + \lim_{M\to\infty} \frac{1}{M} \sum_{N<M} e_N \geq \frac{w}{h},$$

where w is the number of wraps in one doubling-residue cycle and $u_N = ve_N + (p2^N \bmod v)$. If $(p, v) = 1$, then $w \geq 1$.

The exact excess-mean identity is the shifted excess-mean theorem; see also Proposition B.8.

Corollary 4.8 (dyadic reciprocal-mass consequence)**.** *Under the hypotheses of Theorem 4.4, if $v = 1$ and A is infinite, then either $\rho(A)$ diverges or $\rho(A) > 1$.*

This is formalised by the dyadic reciprocal-mass alternative; see also Corollary B.9.

These are four independent necessary consequences of the common rationality premise, not steps in a proof chain and not a contradiction. Appendices B.1–B.2 supply the common recurrence and Boolean certificate; Appendices B.3 and B.4 supply the independent filters. No corollary above is used to prove another.

Corollary 4.9 (exact Boolean–Möbius obstruction for universal #257)**.** *The universal assertion in Erdős #257 is equivalent to the following orbit statement: for every rational number p/q, every Boolean–Möbius carry certificate for p/q reconstructs a finite support. Equivalently, the universal assertion fails exactly when one such certificate reconstructs an infinite support.*

Here a certificate is the positive, square-root-bounded integral orbit with divisibility and Boolean Möbius conditions stated in Theorem B.4. That theorem is the Lean-checked fraction-by-fraction equivalence between supports and carry certificates; the corollary merely quantifies it over p/q and separates finite from infinite reconstructed supports. This is an exact obstruction, not a finite algorithm: the orbit condition and the support it reconstructs are infinite objects.

4.5 The target $1/2$

Let

$$\mathcal{S}(x) = \{n \ge 1 : \varepsilon_n(x) = 0\}$$

be the set of exponents omitted by the greedy expansion.

A worked start. The greedy expansion of $1/2$ begins: $w_1 = 1 > \frac{1}{2}$, so rank 1 is skipped; $w_2 = \frac{1}{3}$ is taken, leaving $\frac{1}{6}$; $w_3 = \frac{1}{7}$ is taken, leaving $\frac{1}{42}$; ranks $4, 5$ are skipped because $w_4 = \frac{1}{15}$ and $w_5 = \frac{1}{31}$ exceed $\frac{1}{42}$; and $w_6 = \frac{1}{63}$, $w_7 = \frac{1}{127}$ are taken, leaving

$$\frac{1}{2} - \left(\frac{1}{3} + \frac{1}{7} + \frac{1}{63} + \frac{1}{127}\right) = \frac{1}{16002}.$$

The next take is at rank 14, and through rank 26 the expansion takes exactly the ranks $2, 3, 6, 7, 14, 20, 21, 26$ (the kernel-checked rank-26 prefix cited in Appendix A.4). Membership of $1/2$ in $\mathcal{A}$ asks whether the complementary skips recur beyond every bound. A *last* skip is precisely a skip whose residual lies strictly above the whole remaining tail (Appendix A.4.6), and Theorem 4.14 below identifies nonmembership with the existence of one.

Definition 4.10 (finite greedy seam and terminal bit)**.** For an integer $s \ge 6$, set

$$q_{s,d} = \left\lfloor \frac{4^s}{2^d - 1} \right\rfloor \quad (2 \le d < s), \qquad K_s = 2^{2s-1} - 2^s.$$

Starting from $R_{s,2} = K_s$, define successively, for $2 \le d < s$,

$$\beta_{s,d} = \begin{cases} 1, & q_{s,d} \le R_{s,d}, \\ 0, & q_{s,d} > R_{s,d}, \end{cases} \qquad R_{s,d+1} = R_{s,d} - \beta_{s,d} q_{s,d}.$$

The Boolean word $(\beta_{s,2}, \ldots, \beta_{s,s-1})$ is the *finite greedy seam at row s*, and $\tau_s = \beta_{s,s-1}$ is its terminal bit.

Here $q_{s,d} = \lfloor 4^s w_d \rfloor$ is the weight w_d at resolution 4^{-s}, and $K_s = 4^s \cdot \frac{1}{2} - 2^s$ is the target $1/2$ at the same resolution, lowered by a guard of 2^s; the word β records which weights this integer greedy run takes. The seam is thus an exact integer coordinate model of the real greedy process. Appendix A.4 develops the equivalence with final skips and the unresolved transition cells; the finite coordinate changes in Appendix A.5 do not strengthen that equivalence.

Theorem 4.11 (exact half-membership classification)**.** *The following are equivalent:*

(i) $\frac{1}{2} \in \mathcal{A}$;

(ii) $\mathcal{S}(1/2)$ *is infinite*;

(iii) $\mathcal{S}(1/2)$ *has no largest element*;

(iv) *for every N there is $s \ge \max\{N, 6\}$ with $\tau_s = 0$.*

The omitted-set equivalence is the infinite-skip theorem; the terminal forms are the unbounded-terminal theorem and no-final-skip theorem.

Proposition 4.12 (finite half-value exclusion)**.** *No finite normalised support has value $1/2$.*

This is the finite-support theorem.

Corollary 4.13 (counterexample from half-membership)**.** *Any one of the equivalent conditions in Theorem 4.11 produces an infinite support $A \subseteq \mathbb{N}_{>0}$ with $X_A = 1/2$, and therefore refutes the universal form of Erdős #257.*

This combines the unique greedy support from Theorem 4.3 with Proposition 4.12. A tempting signed-to-Boolean shortcut also fails: selecting precisely the negative Möbius indices overshoots $1/2$ by at least $1/63$. Proposition B.2 gives the exact sign-separation identity.

Section consequence. Theorem 4.11 replaces a search over arbitrary supports by one canonical quantifier question: does the greedy orbit of $1/2$ skip cofinally? It does not answer that question. The next subsection assumes a possible final skip and identifies exactly what would be required to exclude it.

4.6 The final-skip reduction

For each $s \geq 6$, let

$$G_s = \{d : 2 \leq d < s,\ \beta_{s,d} = 1\}.$$

Let H_s be the unique subset of $\{2, \dots, s-1\}$ for which $\sum_{d \in H_s} q_{s,d}$ is least among the subset sums strictly exceeding K_s. The transition from row s to row $s+1$ is called *upper*, *middle*, or *right* according as

$$G_{s+1} = H_s, \qquad G_{s+1} = G_s, \qquad G_{s+1} = G_s \cup \{s\},$$

respectively. Thus a final skipped exponent D is an upper or middle transition followed only by right transitions. The explicit row-13 computation gives a false terminal bit there. Consequently, if the seam is eventually right, its last false terminal row automatically satisfies $D \geq 13$; the lower bound is not an omitted range of cases. This is the last-false row theorem.

At a middle transition $D \geq 13$, put

$$A_D = G_D \cup \{D\}, \qquad P_D = 2 f_{G_D}(2D+1) + f_{G_D}(2D+2),$$

and write

$$C_D = 4R_{D,D} - P_D - 4 \in \mathbb{Z}, \qquad \Theta_D = \sum_{j \geq 1} \frac{f_{A_D}(2D+2+j)}{2^j}.$$

Here C_D is the integer transition carry and Θ_D its complete future divisor-incidence tail. Their comparison has an exact meaning: by the residual identity of Appendix A.4.7,

$$\frac{1}{2} - \sum_{a \in A_D} w_a = 2^{-(2D+2)} \big(C_D - \Theta_D \big),$$

so the inequality $\Theta_D < C_D$ says exactly that the augmented support A_D falls short of $\frac{1}{2}$.

For a finite set $u \subseteq \{1, \dots, d\}$, write $V(u) = \sum_{n \in u} w_n$. Call (u, d) a *fatal half-gap* if

$$V(u) + T_{d+1} < \frac{1}{2} < V(u) + w_{d+1}.$$

Such a pair certifies that no support agreeing with u through rank d can represent $1/2$.

Theorem 4.14 (final-skip classification)**.** *Nonmembership of $1/2$ in $\mathcal{A}$ is equivalent to each of the following:*

(i) *the integer seam recurrence is eventually right;*

(ii) *a fatal half-gap exists;*

(iii) *the real greedy expansion of $1/2$ has a final skipped exponent.*

The three equivalent nonmembership criteria are formalised in fatal-gap classification, nonmembership classification, and final-skip classification.

Proposition 4.15 (excluded final-skip branches)**.** *At a final skipped exponent $D \geq 13$, the upper successor is impossible. On the middle branch the value $C_D = -3$ is impossible. Among the three exceptional dyadic cells identified by the two-sided induction, only $C_D = -2$ and $C_D = -1$ remain unresolved.*

The upper and $C_D = -3$ exclusions are the upper-branch exclusion and middle--3 exclusion. The three exceptional dyadic cells are identified by the three-cell classification.

Corollary 4.16 (tail-dominance membership criterion)**.** *If every middle transition at a row $D \geq 13$ with $C_D \neq -3$ satisfies*

$$\Theta_D < C_D, \tag{4.2}$$

then no final skipped exponent exists, and hence $1/2 \in \mathcal{A}$.

This implication is the tail-dominance theorem. Because $\Theta_D \geq 0$, the hypothesis of this corollary already rules out every actual middle transition with a negative coordinate other than the locally discharged value -3. Thus the single displayed hypothesis contains two different requirements: exclusion of the unresolved cells $C_D \in \{-2, -1\}$, and the future-tail bound (4.2) on the remaining nonnegative coordinates. Neither requirement is proved for the greedy orbit.

Logical status of the final-skip argument. The proved levels of the argument are

$$\begin{aligned} \tfrac{1}{2} \notin \mathcal{A} &\iff \text{a final skipped exponent exists} && \text{[proved]}, \\ \text{final upper, or final middle with } C_D = -3 &\implies \perp && \text{[proved]}. \end{aligned}$$

These lines do not imply membership. The unproved hypothesis has two parts: exclude all actual middle transitions with $C_D \in \{-2, -1\}$, and prove (4.2) at every remaining non-(-3) middle transition. Together they establish the global hypothesis of Corollary 4.16; only then does Corollary 4.13 produce an infinite rational-valued support and a counterexample to the universal #257 statement.

4.7 What remains open

The two directions remain separate.

(a) The universal problem is to prove $X_A \notin \mathbb{Q}$ for every infinite $A \subseteq \mathbb{N}_{>0}$. Theorem 4.2 proves several families, while Theorem 4.4 and its four corollaries constrain a hypothetical rational support. Neither route supplies the universal quantifier.

(b) The counterexample problem is to prove $1/2 \in \mathcal{A}$. Theorem 4.11 identifies this with infinitely many greedy skips, while Theorem 4.14 identifies nonmembership with a final skip. Proposition 4.15 removes the final upper and middle-(-3) branches. Corollary 4.16 would finish the argument if every actual middle transition avoided $C_D = -2, -1$ and satisfied (4.2) in every remaining non-(-3) cell. Neither global obligation is discharged.

Appendix A records the exact local reductions, finite certificates, counterexamples to stronger local assertions, and closed strategy branches supporting this conclusion. They refine the obstruction; they do not constitute a second theorem spine.

5 The totient constant S (Erdős #249)

Whether

$$S = \sum_{n \geq 1} \frac{\varphi(n)}{2^n}$$

is irrational remains open. The argument has four logically distinct layers. First, a finite Farey computation excludes every rational denominator up to an explicit bound. Second, irrationality is characterised exactly by the existence of finite certificates against every possible eventual binary period. Third, many such certificates are verified at bounded parameters. Fourth, and still missing, is a theorem supplying them at unbounded parameters. Only the fourth layer would settle the problem.

The main section states this theorem spine. Appendix C.1 gives equivalent cofinal coordinates; Appendix C.2 separates fixed-scale mechanisms from the conditional cofinal criteria that consume them; and Appendices C.3–C.4 record necessary inputs, scoped no-go results, and conditional rank routes. Only the first is an exact reformulation of the open fourth layer.

5.1 Unconditional: no small denominator

Theorem 5.1 (denominator exclusion)**.** *Set*

$$Q_0 := 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \approx 7.96 \times 10^{34}.$$

For every integer p and every q with $1 \le q \le Q_0$,

$$S \neq \frac{p}{q}.$$

Equivalently, if S is rational then its reduced denominator exceeds that bound.

Formalised as the record denominator exclusion. The bound is reached by a ladder of finite, elementary exclusions,

$$4838 \;\to\; 4\,194\,304 = 2^{22} \;\to\; 2.49 \times 10^{17}\ (\text{Farey } K{=}120) \;\to\; 7.96 \times 10^{34}\ (\text{Farey } K{=}240),$$

each a mediant/Farey argument that traps any candidate m/q inside a forbidden gap. The mediant framework is historically associated with Farey's 1816 note on vulgar fractions [10]; the finite exclusions themselves are Lean-checked arguments in the formal-source checkpoint. See denominator exclusion through 4838 for the first rung and 120-term Farey bound, 240-term Farey bound for the two Farey windows.

This is an explicit lower bound on the denominator of any rational representation, not a proof of irrationality. A rational with a larger denominator is not excluded by it.

Remark (one bound, four representations)*.* Because the ladder identities of Section 3 are equalities, the same finite record transfers verbatim to three other representations of the constant: to the positive lift $\sum_d \alpha(d)/(2^d-1)$ (positive-lift denominator transfer), to the signed Möbius-square constant $T = \sum_d \mu(d)/(2^d-1)^2$ at half the bound (signed-square denominator transfer), and to the coprimality probability of Section 5.2 at half the bound. Half appears because $S = \frac{1}{2} + T$ turns a denominator d for T into $2d$ for S.

5.2 The geometric picture: S as coprime-pair mass

Let X, Y be independent fair-coin waiting times, $\Pr(X = n) = 2^{-n}$ for $n \ge 1$. Then $\Pr(d \mid X) = 1/(2^d - 1)$, so the Lambert transform is a divisor calculus of X: $L(f) = \mathbb{E}[(f * \zeta)(X)]$. The rungs read as $L(\mu) = \Pr(X = 1) = 1/2$, $L(\varphi) = \mathbb{E}[X] = 2$, $L(\mathbf{1}) = \mathbb{E}[\tau(X)] = E$, and $L(\alpha) = \mathbb{E}[\varphi(X)] = S$.

Counting visible lattice points (a, b) with $a + b = n$, $a > 0$, $\gcd(a, b) = 1$ gives exactly $\varphi(n)$, uniformly in n, so S is itself a coprime-pair mass.

Proposition 5.2 (coprimality probability)**.**

$$S \;=\; \tfrac{1}{2} \;+\; \Pr\bigl(\gcd(X, Y) = 1\bigr) \;=\; \tfrac{1}{2} \;+\; \sum_{\substack{a,b\ge 1\\ \gcd(a,b)=1}} 2^{-(a+b)}.$$

Formalised: coprime-pair representation, with the underlying lattice identity at visible-lattice-point identity. Base 2 is the one point where $\Pr(X = a)\Pr(Y = b) = 2^{-(a+b)}$ needs no normalising constant, so the totient sum and the coprimality probability line up with no boundary correction. In this reading, Erdős #249 asks whether two independent fair-coin waiting times are coprime with irrational probability. After reducing (X, Y) by their gcd, the same distribution has an exact Stern–Brocot cylinder decomposition. Appendix D.3 states the resulting probability law and the sharp Fibonacci stability of its alternating runs. These are unconditional structural theorems inside one representation of S; they do not supply the denominator survival needed for irrationality.

5.3 Exact certificate characterisation

The reduction uses only the elementary bound $\varphi(n) \le n$, geometric tail estimates, and finite integer arithmetic. For $t \ge 1$, write $M_t = \operatorname{lcm}(1, \dots, t)$.

Why tail differences detect rationality. Since $2^N S = \sum_{n \le N} \varphi(n)\, 2^{N-n} + R_N$, each scaled tail R_N is $2^N S$ minus an integer. Suppose $S = p/q$ with $q = 2^c v$ and v odd. For every multiple h of the multiplicative order of 2 modulo v and every $N \ge c$,

$$R_{N+h} - R_N = \frac{2^N\left(2^h - 1\right)p}{q} - k \in \mathbb{Z}$$

for some integer k, because $v \mid 2^h - 1$ and $2^c \mid 2^N$. Rationality therefore supplies one period h all of whose tail differences at large N are integral; conversely, a single integral positive-shift difference already forces S to be rational. A finite certificate refutes one candidate integrality, and refuting every period at cofinally many N refutes rationality outright. This is the mechanism behind the equivalences of this subsection.

Definition 5.3 (tail differences and finite certificates)**.** For $N \ge 0$, define the scaled tail

$$R_N \;=\; \sum_{m \ge 1} \frac{\varphi(N + m)}{2^m}.$$

For $h \in \mathbb{N}_{>0}$ and $N, L \in \mathbb{N}$, define

$$\Delta_{h,N,L} = \sum_{j=0}^{L-1} \bigl(\varphi(N + h + 1 + j) - \varphi(N + 1 + j)\bigr)\, 2^{\,L-1-j} \in \mathbb{Z},$$

with the sum empty when $L = 0$, and say that (h, N, L) is a *finite certificate* when

$$\mathsf{Sep}(h, N, L): \quad N + h + L + 2 \;<\; \left(\Delta_{h,N,L} \bmod 2^L\right) \;<\; 2^L - (N + h + L + 2).$$

The integer $\Delta_{h,N,L}$ contains the first L binary places of $R_{N+h} - R_N$; the two margins bound the omitted tail using $\varphi(n) \le n$. Thus each instance of the predicate is decidable by finite integer arithmetic, while its conclusion concerns the infinite tail difference. The depth-floor theorem implies that a successful instance has $L > 0$, even though the formal existential quantifier ranges over all of $\mathbb{N}$. Heuristically, the two margins together occupy a proportion of about $2(N{+}h{+}L{+}2)/2^L$ of the modulus, which is small as soon as 2^L is large compared with $N + h + L$; a certificate can fail only when the residue lies exceptionally close to 0 or 2^L, that is, when the tail difference is exceptionally close to an integer. Nothing in this paper converts that observation into the required cofinal supply.

Proposition 5.4 (certificate completeness)**.** *For every $h \ge 1$ and $N \ge 0$,*

$$\bigl(\exists L,\ \mathsf{Sep}(h, N, L)\bigr) \quad\Longleftrightarrow\quad R_{N+h} - R_N \notin \mathbb{Z}.$$

This is the certificate-completeness equivalence. Thus the finite predicate loses no information at a fixed pair (h, N). The certificate depth-floor lemma forces every witness to have positive depth. Searching $L = 0, 1, 2, \dots$ is therefore a complete positive semi-decision procedure for non-integrality at fixed (h, N): it halts exactly when the difference is non-integral. When the difference is integral, completeness says that no depth succeeds, so this search does not halt. It is not a two-sided decision procedure for an infinite tail.

Theorem 5.5 (pointwise certificate characterisation)**.** *The following are equivalent:*

$$\begin{aligned}&S \notin \mathbb{Q},\\ &\forall\, h \geq 1\ \forall\, N \geq 0, \qquad R_{N+h} - R_N \notin \mathbb{Z},\\ &\forall\, h \geq 1\ \forall\, N \geq 0\ \exists\, L, \qquad \mathsf{Sep}(h, N, L).\end{aligned}$$

Formalised: all-differences equivalence, and pointwise certificate equivalence. The converse direction uses the fact that integrality of even one positive-shift tail difference forces S to be rational (integral-difference rationality criterion).

Theorem 5.6 (exact tail-period characterisation)**.** *S is irrational if and only if, for every period $h \geq 1$ and every threshold N_0, there exist $N \geq N_0$ and L with $\mathsf{Sep}(h, N, L)$.*

Formalised as the certificate-supply equivalence. Proposition 5.4 supplies the fixed-parameter equivalence; Theorem 5.5 assembles all fixed pairs; and Theorem 5.6 replaces pointwise placement by the equivalent cofinal placement quantifier.

Every multiple of an eventual period is again a period, and every fixed h divides M_t for all $t \geq h$. The least common multiple therefore provides an exact one-parameter form: the single diagonal family (M_t, M_t) tests every candidate period at once.

Theorem 5.7 (exact diagonal characterisation)**.** *The following are equivalent:*

$$\begin{aligned}&S \notin \mathbb{Q};\\ &\forall\, t_0 \in \mathbb{N}\ \exists\, t \geq t_0\ \exists\, L, \qquad \mathsf{Sep}(M_t, M_t, L).\end{aligned}$$

The complete equivalence is the exact lcm-diagonal characterisation. Its reverse implication reuses the diagonal-supply theorem, while its forward implication uses Theorem 5.5 at the pair (M_{t_0}, M_{t_0}). The theorem replaces the two free parameters h, N by (M_t, M_t); producing that cofinal one-parameter family is still open.

Example 5.8 (verified finite range)**.** Lean checks:

- $\mathsf{Sep}(h, 12, 16)$ for every $1 \leq h \leq 8$;
- a fixed-window certificate for every $1 \leq h \leq 16$;
- diagonal certificates $\mathsf{Sep}(M_t, M_t, L)$ at $t = 1$ and every lcm-height change endpoint $2 \leq t \leq 64$ (that is, $M_{t-1} < M_t$), namely

  $$t \in \{1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32, 37, 41, 43, 47, 49, 53, 59, 61, 64\};$$
- a finite joint certificate refuting cone flatness at a tabulated family of vertices.

The four validation classes are formalised, in order, by the small-window certificate family, periods through sixteen, 28 imported diagonal certificates through scale 64, and cone-cell certificate table.

For example, at $(h, N, L) = (1, 12, 16)$,

$$\Delta_{1,12,16} = -143140, \qquad \Delta_{1,12,16} \bmod 2^{16} = 53468,$$

while the omitted-tail allowance is $12 + 1 + 16 + 2 = 31$. Hence

$$31 < 53468 < 2^{16} - 31,$$

which is exactly $\mathsf{Sep}(1, 12, 16)$ and therefore proves $R_{13} - R_{12} \notin \mathbb{Z}$. Every row above has this exact status: it proves a particular non-integral tail difference, not an unbounded supply.

5.4 What remains open

By Theorem 5.6, the unresolved statement is precisely

$$\forall h \geq 1 \; \forall N_0 \geq 0 \; \exists N \geq N_0 \; \exists L \geq 1, \qquad \mathsf{Sep}(h, N, L). \tag{5.1}$$

The diagonal condition in Theorem 5.7 is an equivalent one-parameter form of this statement. The verified scales in Example 5.8 form a finite set and establish neither cofinal quantifier pattern. No theorem in the formal source proves (5.1); consequently no irrationality claim for S is made here.

Appendix C.1 records the exact reformulations of this open obligation. Appendix C.2 records fixed-scale enclosures, projections, and certificates, then states separately the conditional cofinal criteria that consume them. Appendices C.3–C.4 delimit candidate unbounded routes. None supplies the missing cofinal quantifier.

6 The shared finite-to-cofinal bottleneck

This section makes one comparison between the two open spines; it introduces no additional mathematical claim and is not a further proof step. Both use complete finite predicates, but their cofinal conclusions point in opposite directions:

$$\begin{aligned} S \notin \mathbb{Q} \quad &\iff \quad \mathsf{Sep}(h, N, L) \text{ occurs cofinally for every } h, \\ \frac{1}{2} \in \mathcal{A} \quad &\iff \quad \tau_s = 0 \text{ occurs cofinally.} \end{aligned}$$

For S, Proposition 5.4 supplies fixed completeness, Theorem 5.5 gives the pointwise global form, and Theorem 5.6 gives the displayed cofinal form. What is missing is a theorem producing finite refuters without bound. For $1/2$, Theorem 4.11 already identifies cofinal terminal zeros with membership; what is missing is a theorem producing those zeros. The classical full-support theorem is the closed comparison case because its Bertrand–CRT and averaging argument constructs the required witnesses uniformly. The coordinate identities and rational-support filters used elsewhere in the paper do not alter either missing quantifier.

Thus the comparison also gives a criterion for further progress. On the #249 side, a new local identity affects the endpoint only if it produces or rules out Sep-certificates at unbounded parameters. On the half-value side of #257, it must produce cofinal terminal zeros or prove that the greedy seam is eventually right. Further fixed-scale computation, by itself, changes neither open statement.

7 Formal verification and reproducibility

This section records proof authority and reproducibility; it does not add a mathematical conclusion to either theorem spine.

Proof authority and source identity. The Lean 4 proof assistant [19] and Mathlib library [20] check the named results. The source, exact informal-to-formal links, and build instructions are available in the public repository. This draft names formal-source revision `c2c5c3f0151d`, but that revision must be present in the public repository before release. Until the source commit is pushed and the source-coordinate checker is rerun, the blue links are navigation text rather than a reproducible proof-authority route. The final published revision must also pin the Lean and Mathlib versions; the intended toolchain is Lean `v4.29.1`.

Reproduction commands and artefact identities. For release, a cold clone at the published revision must pass `lake build`; the release-validation command must then check the claim registry, source coordinates, manuscript anchors, projections, trust guards, the rendered PDF, and the immutable committed reference. This draft does not carry that completed release receipt. The machine-readable corpus descriptor records SHA-256 identities for the claim registry, declaration atlas, manuscript source, rendered PDF, and toolchain projections, together with their regeneration commands. No hardware-independent build time is claimed; elapsed time depends on the Lean cache and machine, and is not part of the mathematical claim.

Computational and expository boundary. Every declaration is elaborated by Lean and every proof term is checked by the kernel. Finite calculations use kernel-reducible procedures such as `decide`, `norm_num`, or explicit certificate proofs. In particular, `finalMiddleTwentySixPhaseSurvivors_card` uses `decide` under a finite declaration-local recursion-depth budget; its axiom audit contains no `Lean.trustCompiler` and no proof placeholder. Generated certificate shards are Lean source; no external certificate dataset, Python release script, JSON projection, or this manuscript is proof authority. Appendix E explains how each displayed result is connected to its exact formal statement.

Section consequence. After the source commit is published and the committed-ref release checks pass, the linked declarations and their checked proof terms will certify the stated proved, conditional, finite, and equivalence results at that revision. That verification does not change the mathematical status of the unresolved statements: the open problems collected in Section 8 remain open.

8 Exact frontiers and further questions

Section 2 gives the exact reductions. Here we record two concrete sufficient criteria and then state the unresolved questions. Erdős #249 has one open obligation in two equivalent forms. Erdős #257 has a universal statement and a counterexample route pointing in opposite directions: half-value membership would refute the universal statement, while nonmembership would close only that route.

8.1 Two concrete scalar criteria

Two scalar conditions give concrete sufficient routes toward the two open directions. Both are exact arithmetic statements; neither is proved at the unbounded range needed for its final conclusion.

For #249, retain $M_t = \mathrm{lcm}(1, \dots, t)$ and write

$$P_L(N) = \sum_{j=0}^{L-1} \varphi(N + 1 + j)\, 2^{L-1-j}.$$

The canonical adjacent-suffix depth and its residue are

$$m_t = \lfloor \log_2 M_t \rfloor + 10, \qquad d_t = \big(P_{m_t}(2M_t) - P_{m_t}(M_t)\big) \bmod 2^{m_t},$$

where $0 \le d_t < 2^{m_t}$. Define

$$\sigma_t = \min\big(d_t - 2^{m_t-5}, (2^{m_t} - 2^{m_t-5}) - d_t\big) \in \mathbb{Z}.$$

Proposition 8.1 (canonical central-band equivalence)**.** *For every $t \ge 1$, the canonical residue lies in the central band $[2^{m_t-5}, 2^{m_t} - 2^{m_t-5}]$ if and only if $\sigma_t \ge 0$.*

This fixed-scale equivalence and its cofinal strict-jump form are the canonical central-band equivalence and strict-jump slack equivalence.

Corollary 8.2 (cofinal strict-jump slack criterion)**.** *If, for every t_0, there is a $t \geq \max\{3, t_0\}$ such that*

$$M_t < M_{t+1} \qquad \textit{and} \qquad \sigma_t \geq 0,$$

then S is irrational.

This implication is the cofinal-slack irrationality criterion. The proposition identifies equivalent scalar forms; the corollary is one-way and does not prove that σ_t is nonnegative cofinally.

For #257, write r_n for the exact greedy residual of the target $1/2$ and put

$$u_n = 4^n(2r_n - 2^{-n}).$$

The second-channel condition is the rational inequality

$$\frac{1}{6} + \frac{37}{56}2^{-n} \leq \left| u_n - \frac{1}{3} \right|.$$

Proposition 8.3 (finite second-channel verification)**.** *The second-channel condition is decidable over $\mathbb{Q}$ and holds for $1 \leq n \leq 6$.*

This is the finite second-channel verification.

Corollary 8.4 (second-channel membership criterion)**.** *If the second-channel condition holds for every $n \geq 7$, then $1/2$ belongs to the Mersenne achievement set $\mathcal{A}$.*

This implication is the from-seven membership criterion. The membership conclusion is itself exact:

$$1/2 \in \mathcal{A} \iff \text{the canonical greedy orbit omits infinitely many exponents.}$$

This is formalised at half-membership–infinite-skip equivalence. The forward implication uses the full-support irrationality theorem; the reverse implication uses the absorbing fatal-state criterion. Thus the second-channel hypothesis is one sufficient route to the infinite-skip condition. It does not prove that condition, and membership alone is not the universal #257 theorem. The finite proposition supplies no case of the missing range $n \geq 7$.

Dependency direction. The two criteria have the proved logical form

$$\begin{array}{c}\text{cofinal nonnegative slack}\\ \text{at strict lcm jumps}\\ \Downarrow\\ \text{unbounded \#249 certificate supply}\\ \Downarrow\\ S \notin \mathbb{Q}\end{array} \qquad\qquad \begin{array}{c}\text{second-channel separation}\\ \text{for every } n \geq 7\\ \Downarrow\\ \dfrac{1}{2} \in \mathcal{A}\\ \Updownarrow\\ \mathcal{S}(1/2) \text{ is infinite.}\end{array}$$

Neither left-hand premise is proved. The verified range $1 \leq n \leq 6$ closes only the finite prefix of the second criterion; it gives no part of the missing universal range $n \geq 7$. The first conclusion settles #249, whereas the second would refute, not prove, the universal #257 statement.

8.2 The unresolved statements

Erdős #249: one obligation in two equivalent forms.

Problem 8.5 (Erdős #249)**.** Prove that

$$S = \sum_{n \geq 1} \frac{\varphi(n)}{2^n}$$

is irrational.

By Theorem 5.6, the preceding problem is equivalent to the following exact quantifier statement.

Problem 8.6 (unbounded certificate supply)**.** Prove (5.1). Equivalently, prove the lcm-diagonal supply in Theorem 5.7. The finite scales in Example 5.8 do not establish either cofinal form.

These are not two independent open problems: Theorem 5.6 and Theorem 5.7 identify the second as an exact certificate form of the first.

Erdős #257: universal statement and counterexample route.

Problem 8.7 (universal Erdős #257)**.** Prove that

$$\sum_{n \in A} \frac{1}{2^n - 1}$$

is irrational for every infinite set $A \subseteq \mathbb{N}$, rather than only for the support families in Theorem 4.2.

Problem 8.8 (the half-value counterexample route)**.** Decide whether $1/2 \in \mathcal{A}$. By Theorem 4.11, this is equivalent to the canonical greedy orbit omitting infinitely many exponents. A positive answer would give an infinite support of rational value $1/2$ and refute the universal #257 statement; a negative answer would close this distinguished counterexample route only.

How the frontier has changed. The results above change the form, not the status, of the open problems. For #249, the unresolved content is one unbounded certificate-supply theorem, with the pointwise and lcm-diagonal forms proved equivalent to irrationality. For the half-value route to #257, the search over arbitrary supports has been replaced by one canonical greedy-orbit question. In the present final-skip argument that question is localised further to two obligations: exclude the middle cells $-2, -1$, and prove the future-tail inequality at the remaining middle transitions. A cofinal #249 certificate producer would prove #249; a cofinal terminal-zero producer would refute universal #257; and a proof of eventual right behaviour would close only the half-value route.

Statements and declarations

Artefact and data availability. The archived formal-source revision contains the Lean sources, pinned toolchain, library manifest, and generated certificate data used in the verification. Repository metadata and this manuscript provide navigation rather than proof authority; Section 7 gives the verification route.

Use of AI assistance. Large-language-model assistants were used during development for proof drafting, refactoring, and prose editing. Every registered formal claim is linked to a declaration in the archived source revision. Lean checks each proof term against the pinned library; the project contains no proof placeholders or project-defined axioms.

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Acknowledgements. The problem numbering and status follow the Erdős Problems catalogue maintained by Thomas Bloom [21].

A Technical reductions for the half-value branch

This appendix records the local arithmetic used to reach Theorems 4.11 and 4.14, Proposition 4.15, and Corollary 4.16. Its statements fall into four classes: exact coordinate identities, conditional compactness criteria, bounded finite certificates, and counterexamples to tempting local inductions. None changes the two open conclusions at the end of Section 4. The appendix is included so that the hypotheses in the main reduction can be audited without turning the main theorem section into a chronology of the formal development.

The order is by mathematical use. Appendix A.1 localises one possible unsafe skip; Appendix A.2 records conditional structured-support criteria; Appendix A.3 collects the finite-window and compactness implications; Appendix A.4 develops the equivalent final-skip coordinates; and Appendix A.5 records verified finite instances, exact coordinate translations, counterexamples, and closed strategy branches. This is a dependency map, not a second route to a conclusion.

The logical spine is

$$\begin{array}{rcl} \text{cofinal half-value witnesses from A.2–A.3} & \Longrightarrow & \tfrac{1}{2} \in \mathcal{A}, \\ \tfrac{1}{2} \in \mathcal{A} & \Longrightarrow & \text{an infinite rational-valued support}, \\ \text{a final greedy skip} & \Longleftrightarrow & \tfrac{1}{2} \notin \mathcal{A}, \\ \text{the A.4 residual cell exclusions and tail bound} & \Longrightarrow & \text{no final skip}, \\ \text{no final skip} & \Longrightarrow & \tfrac{1}{2} \in \mathcal{A}. \end{array}$$

The first two and last two lines are conditional implications; the middle line is an exact classification. Appendix A.1 only localises one transition configuration, while Appendix A.5 supplies bounded evidence, coordinate changes, and no-go results. Neither creates a further implication to an open endpoint.

Throughout this appendix, a *take* (respectively a *skip*) at rank n is the greedy decision $\varepsilon_n = 1$ (respectively $\varepsilon_n = 0$); a *seam* is an adjacent pair of finite greedy words at a branch boundary; a *rewind* is the corresponding inverse finite recurrence; the *transition carry* is the integer coordinate appearing in the exact residual identity; the *remainder* of a row is its integer greedy residual; and the *pulse* of a rank at row s counts its divisor incidences at $2s+1$ and $2s+2$, the first counting twice, the pulse of a word being the sum over its ranks. These terms denote mathematical objects, not stages of a proof search. Module-local coordinates not listed here (for example the centred, lazy, or first-wrap variants) are defined in the linked formal statements.

A.1 Exact localiser: a band formula for the final skip between two takes

This subsection has one diagnostic role: it gives the exact unsafe interval for a specified take–skip–take configuration. It helps interpret the transition geometry used in the final-skip analysis, but it is neither an orbit-level exclusion nor a premise that produces half-value membership.

One local configuration admits a simple exact description in full generality. Suppose a take at rank b is followed by the next take at $c > b + 1$. Put $q = 2^b - 1$, $m = 2^{c-1}$, and write the residual immediately before the first take as $1/R$. When $0 < R < q$, the final intervening skip is dyadically unsafe if and only if

$$\frac{q(m-1)}{q+m-1} < R < \frac{qm}{q+m}.$$

The interval has exact width
$$\frac{q^2}{(q+m)(q+m-1)}.$$
Formalised: the general unsafe-band equivalence and the general band-width formula.

For a single intervening skip, $c = b+2$ and $m = 2q+2$. The band becomes
$$\frac{q(2q+1)}{3q+1} < R < \frac{2q(q+1)}{3q+2},$$
with width $q^2/((3q+1)(3q+2)) < 1/9$, entirely between $2q/3$ and $2q/3 + 2/9$. No integral reciprocal lies in this window. In the corresponding positive odd integer formulation, with residual $p/(2D)$, membership in the cleared band forces $p \geq 7$. Formalised: the one-skip band equivalence, the one-skip width formula, the width bound, the integral-reciprocal exclusion, and the odd-coordinate lower bound.

These are localisers, not orbit-level eliminations. They do not show that the greedy orbit of $1/2$ avoids any such band, and they do not identify the two remaining exceptional transition coordinates $-2, -1$ with dyadically unsafe skips.

A.2 Conditional criteria and local inputs for structured supports

The results here have three distinct roles. The bouquet criterion is a conditional irrationality theorem for one structured support family. The seam, strip, and terminal criteria are sufficient hypotheses for $1/2 \in \mathcal{A}$. The dilation, prime-power, and return-carry results are local algebraic inputs that yield neither endpoint without an additional supply hypothesis. Nothing in this subsection is a new unconditional case.

A.2.1 A conditional pairwise-coprime dilation criterion

Let A be a finite-core orthogonal-petal bouquet: outside a finite frame, its elements have the form $c_i p_i$, where the cores c_i divide one fixed integer and the nontrivial petals p_i are pairwise coprime to that frame and to one another, with summable reciprocal mass. The formal construction also contains an explicit infinite example with alternating cores 2 and 3 and squared prime petals; every member of this support has exactly three prime factors with multiplicity.

For such a bouquet, the remaining input is an exact tail-selection predicate: at every positive binary scale it chooses a carry slot with a normalised tail at most 16. Lean proves that this predicate supplies the needed forced-carry certificates and hence irrationality of the base-2 support series. The result is the conditional bouquet irrationality theorem, with the carry bridge at the forced-carry implication. The tail-selection predicate itself is not proved here. Thus this is a conditional reduction for one structured support family, not a further unconditional case of Erdős #257.

A.2.2 Composite dilation defects

For a support divisor count $f_A(x) = \#\{d \in A : d \mid x\}$, multiplication by a composite support element a has an exact correction term. Besides the distinguished divisor a, the new divisors are counted by a finite defect which excludes divisors already present at x. The formal identity is the composite-dilation identity. For a finite-core orthogonal-petal bouquet, the defect at a ray is bounded by the finite exceptional frame plus the divisor count of the petal support: the bouquet defect bound. This records the local correction needed when dilation is not prime. It does not prove a correlation estimate, the sunflower tail selector, or an irrationality statement.

A.2.3 Mixed prime-power layers

The one-prime-power layer has a two-signature extension. Exact p-adic and q-adic layer operators commute, and, for distinct primes with the residual argument coprime to pq, their composite extracts the iterated exact-valuation pullback of the support coefficient. This is formalised by the two-prime layer identity. The singleton support $\{12\}$ gives a checked $(2^2, 3)$ example with value one at residual argument 1: the singleton-support calculation. These are finite coefficient identities. They do not provide a decimation transport, a bounded-Ω conclusion, or an irrationality theorem.

A.2.4 Half-trapping return carries

If rational-linear channels vanish on every relation killed by one scalar evaluation, their evaluation matrix has rank at most one; every nontrivial square minor therefore vanishes. The all-rank determinant statement is the rank-one determinant criterion. For two reverse carries with a common coefficient word, a $1/0$ seam, and a common following overlap, the terminal difference is a power of two times an odd integer. A shared nonnegative Archimedean bound then gives the matching dyadic spacing inequality, formalised at the dyadic spacing inequality. The argument does not produce reverse-carry presentations or an earlier overlapping return; it supplies only the algebraic core of that conditional criterion.

A.2.5 Half-carry reachability

There is also a finite-prefix criterion for the target $1/2$. At an unresolved even seam, the scalar transition reaches every integer terminal carry except the single value $2\delta - c$; this is the exact equivalence the even-seam reachability equivalence. A canonical description of such a seam at every sufficiently large even depth is an unproved hypothesis. Under that hypothesis, the finite admissible words are cofinal, by the cofinal-admissibility implication, and compactness then produces an infinite support with sum exactly $1/2$: the half-value compactness conclusion. Thus the formal result is a conditional reduction. It does not establish the canonical seam supply and does not prove Erdős #257.

Cofinal strip returns. The all-level strip hypothesis can be weakened for the canonical greedy half carry. Its uncentred, binary-scaled form is antitone. Consequently, if it returns to the square-root strip cofinally, then one sufficiently late return bounds every later scaled carry; the strip envelope divided by 2^N tends to zero. The resulting tempered centred carry yields a support selected by the greedy expansion whose series value is $1/2$; the finite-support exclusion makes that support infinite.

Proposition A.1 (cofinal strip-return criterion)**.** *Let G_M be the uncentred integer carry of the canonical greedy expansion of $1/2$. If*

$$\forall N \in \mathbb{N}\ \exists M \geq N, \qquad G_M \leq 2\lfloor\sqrt{M+1}\rfloor + 4,$$

then $1/2 \in \mathcal{A}$, and its canonical greedy support is infinite.

Formalised: scaled-carry decay and the strip-return membership criterion. The cofinal return condition itself is not established. It is weaker than a uniform strip bound and still does not give an unconditional membership theorem.

Terminal-only cofinal approximation. Coherence can be removed entirely from a further version of this reduction. At depth M, take only a normalised finite word whose terminal half carry lies in the square-root strip. Its finite support then has support-series value within

$$\frac{4M+12}{2^M}$$

of $1/2$. If such terminal witnesses occur cofinally, these finite values converge to $1/2$; closedness of the Mersenne achievement set supplies the limit. No control of the earlier rows, and no compatibility between the witnesses, is used.

Proposition A.2 (terminal-only cofinal criterion)**.** *Cofinal normalised terminal witnesses in the square-root strip imply that* $1/2$ *belongs to the Mersenne achievement set, and hence that an infinite support has support-series value* $1/2$*.*

Formalised: the terminal approximation bound, the terminal-strip compactness criterion, and the infinite-support conclusion. The cofinal terminal-witness supply is not proved. In particular, the proposition does not establish membership of $1/2$ unconditionally.

Scaled terminal vanishing. The square-root strip is stronger than the analytic condition used by the terminal-only argument. Let the depths of normalised finite words tend to infinity. If their terminal carries, after division by the corresponding power of 2, tend to zero, then the universal coefficient-tail error also tends to zero. The finite support values therefore converge to $1/2$, and closedness of the achievement set again supplies the limit. The earlier terminal-only strip condition implies this weaker sequential hypothesis.

Proposition A.3 (scaled terminal-vanishing criterion)**.** *A normalised terminal-word sequence with depths tending to infinity and vanishing binary-scaled terminal carries yields an infinite support with support-series value* $1/2$*.*

Formalised: the scaled terminal error bound, the scaled-vanishing compactness criterion, and the scaled-vanishing support conclusion. No such scaled-vanishing sequence is constructed here; the result is a conditional reduction, not an unconditional membership theorem.

A.3 Finite-to-cofinal criteria: windows, seams, and compactness

Every half-value route in this subsection factors as

$$\text{finite witness at depth } N \quad + \quad \text{a cofinal supply of such depths} \quad \Longrightarrow \quad \frac{1}{2} \in \mathcal{A}.$$

The local identities and recurrences describe or advance one finite witness. The compactness statements turn a cofinal supply into membership. No result below proves that supply, so a verified window or seam at one depth is evidence for a possible recurrence, not an unbounded theorem.

A.3.1 Finite shadows of the half cylinder

For an admissible finite word, the residual from $1/2$ is given exactly by its terminal half carry minus the future coefficient tail, scaled by a power of two; see the finite-shadow residual identity. At a first seam, avoiding the integer strip is equivalent to full scalar reachability of that strip: the seam-escape equivalence. The rank-three greedy example is raw dyadically safe while its seam has an in-strip hole, so raw safety alone does not imply this escape. Finally, an explicit frozen-margin hypothesis implies $1/2$ belongs to the Mersenne achievement set, by frozen-margin implication. That hypothesis is unproved; this is a conditional reduction, not a proof of Erdős #257.

A.3.2 Fixed-coefficient rewind

For a finite list of common coefficients, the canonical inverse-parent map has an exact closed form with a power-of-two denominator: the rewind closed form. An interval of width at most that denominator contracts either to one ancestor or to two adjacent ancestors. The singleton alternative is equivalent to the explicit dyadic phase inequality, while the other alternative is a seam pair; see the singleton-or-seam dichotomy. Thus width alone does not justify a singleton conclusion. This is a finite rewind lemma, and supplies neither a seam exclusion nor an Erdős #257 conclusion.

A.3.3 Selected half-carry windows

The selected-window construction keeps an explicit admissible word for each terminal carry in a finite protected interval. A supply of such windows at all sufficiently large levels gives cofinal admissibility and hence an infinite support with sum $1/2$: selected-window cofinality and the selected-window half-value conclusion. The unresolved input is the coherent history at each step: either a common next-row divisor cylinder, or the canonical adjacent seam. The theorem states this input exactly; it does not construct the window supply and does not prove Erdős #257.

A.3.4 A realised rewind criterion

For a realised fixed-coefficient history, the dyadic rewind phase is the complementary binary suffix numeral: the suffix-phase identity. Consequently the singleton criterion is exactly the inequality that this suffix numeral covers the rest of the terminal interval, formalised by the suffix-coverage criterion. This converts the phase condition into support-history coordinates, but does not prove the required coverage inequality globally.

A.3.5 Localised protected seams

The compactness argument needs only one admissible word at cofinally many depths. At a localised one-hole seam, one of the two protected carries 3 and 4 survives; therefore a cofinal supply of realised protected seams implies cofinal admissibility by the protected-seam cofinality implication, and hence an infinite support with sum $1/2$, by compactness. The cofinal realisation supply is not proved.

A.3.6 Combining the two cofinal half-carry criteria

Assume that at each requested scale there is either a selected protected window or a localised realised seam. Each alternative supplies the one admissible word needed for compactness, so this cofinal disjunction implies an infinite support with sum $1/2$: by window-or-seam cofinality and by the window-or-seam half-value conclusion. The cofinal disjunction remains unproved.

A.3.7 Integer-greedy first-wrap reduction

The first-wrap seam has exact truncated Mersenne weights with a quantitative gap-dominance property. For these weights, a small positive subset-sum defect exists exactly when the descending integer-greedy remainder is small and positive; this is formalised by the integer-greedy remainder equivalence. The remaining arithmetic input is an unbounded lower bound for that deterministic remainder sequence. No such lower bound is proved here.

A.3.8 Exact finite recurrence for the greedy seam

The finite seam-word construction realises the abstract adjacent cut by concrete Boolean words and identifies its next word with the integer-greedy choice: the seam-word recurrence. It implements a finite recurrence, not the unbounded remainder estimate.

A.4 Exact final-skip spine and its residual cells

This is the only subsection that feeds the main final-skip reduction directly. It first identifies nonmembership with a final skip and membership with cofinal terminal zeros. It then separates the terminal cells already excluded from the two unresolved middle coordinates and the unresolved future-tail inequality. The later largest-skip, pulse, and reset formulas are conditional ways to attack those residual obligations; they do not add a second conclusion.

A.4.1 Fatal gaps and eventual right tails

For the concrete integer seam words, eventual right extension is equivalent to nonmembership of $1/2$ in the Mersenne achievement set: the eventual-right nonmembership equivalence. The same condition is equivalent to the existence of a finite fatal half gap, by the fatal-gap equivalence, and also to a final skipped exponent in the real greedy orbit, by the final-skip equivalence. Thus this equivalence gives an exact classification of the nonmembership branch. It neither decides whether $1/2$ belongs to the achievement set nor proves the universal Erdős #257 statement.

This is the nonmembership half of the argument summarised earlier: a last greedy skip is equivalent to eventual right extension. The following paragraph records the first exceptional middle case that can be ruled out.

First final-middle cell. The first exceptional middle cell has a separate obstruction: if the orbit extends right after the last skipped rank, then the cell with transition carry -3 makes the lazy cofinite tail strictly smaller than $1/2$ while its centred carry becomes negative. This is impossible by the analytic nonnegativity identity, the middle--3 exclusion. It removes that one conditional terminal configuration; it does not decide the right-tail branch or the remaining two middle cells.

Second final-middle cell. The next cell, with transition carry -2, is not yet eliminated, but its possible phase is sharply constrained. Uniformly for seam rows $s \geq 27$, the greedy decisions through rank 26 select exactly the ranks

$$2, 3, 6, 7, 14, 20, 21, 26,$$

by the exact rank-(26) prefix. Consequently, if a putative last skipped rank $D \geq 27$ lies in this middle cell and is followed by an all-right suffix, then

$$D \bmod 21 \in \{11, 14, 17, 20\}, \qquad D \bmod 10 \neq 8, \qquad D \bmod 13 \neq 11,$$

the class $D \bmod 10 = 7$ cannot pair with $D \bmod 21 = 11$, and the class $D \bmod 13 = 10$ cannot pair with $D \bmod 21 = 11$. The orbit theorem is the final-middle phase sieve. Exactly 412 of the 2730 joint residue classes survive; this separate finite count is the exact survivor count. This is a uniform exact reduction of the -2 cell, not its exclusion; the surviving phases, the -1 cell, and the right-tail branch remain open.

The remaining global reduction is now precise. If every genuine middle transition other than the already discharged $C_D = -3$ cell has transition carry larger than its complete future divisor-incidence tail, then an eventual right seam is impossible and $1/2$ belongs to the Mersenne achievement set: middle-tail implication. The stronger square-root lower-bound hypothesis is also sufficient, square-root sufficient condition. Neither hypothesis is supplied here.

For finite seam prefixes the remaining tail bound sharpens further: the future divisor-incidence tail is bounded by the cardinality of the prefix. Thus the explicit row-scale inequality $\#G_D + \mathrm{belowPulse}_D + 5 < 4\,\mathrm{remainder}_D$ is sufficient for the same half-membership conclusion. The hypothesis is the cardinality escape condition, and its membership consumer is the prefix-cardinality bound; it is enough in turn that every late middle remainder is at least its row, row-scale sufficient condition. These are exact conditional reductions, not proofs of the row bound.

Finite upper-reset band certificates. The companion upper branch has a kernel-checked finite certificate: for every upper transition from rows 13 through 30, every subsequent dyadic danger band is avoided. The certificate shares each exact successor remainder across all its band indices, the verified upper-reset certificate. It is a bounded transition check and does not supply the cofinal band-escape hypothesis.

A.4.2 Positive half-membership classification

The complementary branch has an equally exact seam formulation. Membership of $1/2$ in the Mersenne achievement set is equivalent to the occurrence, beyond every bound, of a successor terminal bit equal to zero, the unbounded terminal-zero equivalence; equivalently, it is equivalent to a cofinal sequence of such terminal zeros, the cofinal terminal-zero equivalence. It is also equivalent to a cofinal integer-seam sequence with skipped ranks tending to infinity, the unbounded skipped-rank equivalence. These equivalences isolate the required cofinal skipped-rank condition; they do not establish that condition.

A.4.3 Largest skipped ranks

At a named largest skipped rank d, the lower seam word and its adjacent upper competitor have an exact late-range gap. In division-free form, the identity is

$$3L + \left(3 \cdot 2^{s+1} + 2 \cdot 4^{s-d} + 4\right) = 3U$$

whenever $2s < 3d$; this is formalised at the exact late-gap identity. The upper and middle successor branches create a terminal largest skipped rank, while the right branch preserves an existing one: the terminal largest-skip branch and right-branch persistence. The first-crossing branch hypothesis required to use this identity globally is not proved.

A.4.4 Boundary-pulse normalisation

Above a late largest skipped rank, every filled suffix rank has zero row pulse, so the word pulse is carried entirely by the lower prefix: the lower-prefix pulse localisation. At the first crossing of the two-thirds boundary there are exactly two cases, $3d = 2s + 1$ and $3d = 2s + 2$, and the skipped rank has pulse respectively two or one: the first-crossing pulse classification. This supplies the incidence normalisation for a transition-carry estimate; it does not establish that estimate.

A.4.5 Largest-skip induction

Assume that at the first crossing a late largest skipped rank either remains late at the next row or forces an upper-or-middle successor branch. This hypothesis propagates a late largest skipped rank from row 14 onward, largest-skip propagation step, and hence yields cofinal skipped ranks and membership of $1/2$ in the Mersenne achievement set, largest-skip membership criterion. The first-crossing hypothesis itself remains unproved.

A.4.6 The fixed-tail survival criterion

A skipped rank is final exactly when its post-decision remainder lies strictly above the complete remaining Mersenne tail: the final-skip criterion. Consequently membership of $1/2$ is equivalent to the pointwise assertion that every actual skipped rank has remainder at most its full future tail, the half-membership survival equivalence. This is an exact local reformulation; the tail-survival inequality is not proved.

A.4.7 The carry identity at a non-right transition

For a finite support aligned with a non-right transition, the residual from $1/2$ is exactly the integer transition carry minus the complete future divisor-incidence tail, scaled by $2^{-(2d+2)}$: transition-carry residual identity. Thus the carry exceeds that tail exactly when the support series lies below $1/2$, and a negative carry places it above $1/2$; positive-carry criterion and negative-carry criterion. An upper carry at most -8 even gives a full Mersenne-gap margin, full-gap carry criterion. These are local criteria, not a proof that the required carries occur.

A.4.8 Quarter-band endpoint cells

For the adjacent seam recurrence, failure of the desired next-row upper bound on the middle branch is exactly one multiple-of-four pulse cell, the middle-branch pulse-cell equivalence. On the terminal-reset branch, the same failure is exactly a constrained $3 \cdot 2^{s-1} + k$ pulse cell, the reset-branch pulse-cell equivalence. These are arithmetic normal forms for the exceptional cells; no avoidance statement for the concrete seam orbit is proved.

A.4.9 Reset-deficit escape

At a late largest skipped rank, a right successor branch pins the integer greedy remainder between an explicit quarter threshold and carry threshold: the right-branch remainder window. Along such a branch the remainder satisfies an exact affine recurrence, the right-branch affine recurrence. Moreover, a doubled-scale deficit forces an upper-or-middle branch within the corresponding number of rows, the deficit escape bound. The remaining input is the deterministic anti-concentration needed to exclude the pinned right-branch window globally.

A.4.10 Alignment of seam and transition carries

At a first-wrap seam, the transition carry of the left boundary support is four times the shifted seam hole, less the explicit paired incidence pulse at the next two rows: seam–carry alignment. This is an exact coordinate change between the seam and transition coordinates; it does not supply a sign or size estimate for that carry.

A.5 Finite evidence, coordinate changes, and closed branches

This subsection has no open-endpoint theorem. Its verified bases and stages are bounded evidence for the cofinal criteria in A.2–A.3; its coordinate coordinate changes translate one finite formulation into another; and its counterexamples and incompatibility theorems close shortcuts that would otherwise be mistaken for global arguments. Each item must therefore be read through the earlier criterion it supports or blocks.

A.5.1 A verified initial selected window

The selected half-carry construction has a checked depth-18 base: twelve Boolean words cover every terminal carry in the strip $[1, 12]$, with all intermediate carries in the corresponding half strip. The reflected table is formalised at the depth-18 table verification; the resulting selected window is defined at the depth-18 selected window. All twelve words share the same restriction at depth 13, giving the first next-row divisor-agreement certificate, the depth-18 divisor-agreement certificate. This finite base does not provide selected windows at unbounded depths.

A.5.2 Rewind-to-seam equivalence

For a selected window with a rewind seam, a two-value coefficient profile on the adjacent base ancestors induces the live next-row profile, the next-row rewind profile. Under the stated strip and buffer inequalities, that profile realises a protected one-hole seam at the next row, the protected-seam realisation. The base-ancestor coefficient drop is an explicit input, not a theorem here.

A.5.3 Selected suffix cylinders

Put

$$H_N = 2\lfloor\sqrt{N}\rfloor + 4.$$

For a Boolean word $a = (a_0, \dots, a_N)$, let

$$f_a(n) = \#\{d \le N : d \mid n,\ a_d = 1\}, \qquad K_0(a) = 1, \qquad K_{r+1}(a) = 2K_r(a) - f_a(r+2).$$

Definition A.4 (selected window and suffix cylinder)**.** A *selected window* of depth N and radius R is a family $(a^{(k)})_{1\le k\le R}$ of Boolean words with $a_0^{(k)} = a_1^{(k)} = 0$ such that

$$1 \le K_{n-1}(a^{(k)}) \le H_n \quad (1 \le n \le N), \qquad K_{N-1}(a^{(k)}) = k.$$

For $M \le N$, define the suffix numeral

$$\nu_{M,N}(a) = \sum_{j=M+1}^{N} a_j\, 2^{N-j}.$$

The window has a *suffix cylinder at cutoff M with endpoint E* if all $a^{(k)}$ have the same prefix through M and

$$\nu_{M,N}(a^{(k)}) + k = E \qquad (1 \le k \le R).$$

A *profiled gap stage* additionally permits one missing carry interval $[g_-, g_+] \subseteq [1, H_N]$: a lower suffix-cylinder family represents $1, \dots, g_- - 1$, an upper family represents $g_+ + 1, \dots, H_N$, and the two families retain fixed consecutive prefixes through the cutoff.

The results below have five different roles. The checked base feeds the ordinary one-row update. At a feedback row there is a dichotomy between a new full stage and a local gap. A separate recurrence advances such a gap when its coefficient inequalities hold. The quarter-band bound is only a necessary obstruction. Finally, the cofinal corollary is the sole step from arbitrarily deep full stages to half-membership.

Proposition A.5 (verified base and one-row cylinder update)**.** *The depth-*18 *selected window has $R = 12$ and a suffix cylinder at cutoff $M = 13$ with endpoint $E = 17$.*

More generally, let a depth-N, radius-R selected window $(a^{(k)})_{1\le k\le R}$ have a suffix cylinder at cutoff $M \le N$ with endpoint E, where $1 \le N$ and $R \le E$. Let $1 \le R' \le H_{N+1}$. If

$$f_{a^{(k)}}(N+1) = C \qquad (1 \le k \le R)$$

and

$$R' + C \le 2R,$$

then the radius-R' successor remains a suffix cylinder, with endpoint

$$E' = 2E - C.$$

The base and endpoint update are the depth-18 cylinder and endpoint recurrence.

Proposition A.6 (feedback-row dichotomy)**.** *Let a full suffix-cylinder stage have depth N, cutoff M, with $1 \le N$, $M+1 \le N$, and*

$$N + 1 = 2(M + 1).$$

At depth $N+1$, either a full stage exists at cutoff $M+1$, or the exceptional carries form a two-sheet profiled gap stage. If $27 \le H_{N+1}$, the residual alternative also realises a protected one-hole seam.

Formalised by the feedback dichotomy and two-family gap alternative.

Proposition A.7 (profiled-gap transition)**.** *Suppose a profiled gap stage has lower and upper next coefficients C_- and C_+, respectively. If*

$$\begin{gathered} 1 \le 2g_- - C_- - 1, \qquad C_+ \le 2g_+, \\ 2g_- - C_- - 1 \le 2g_+ - C_+ \le H_{N+1}, \\ H_{N+1} + C_+ \le 2H_N, \end{gathered}$$

then the stage advances one row with exact child gap

$$[g'_-, g'_+] = [\,2g_- - C_- - 1,\ 2g_+ - C_+\,].$$

At a cutoff boundary $N = 2M+1$, $M \ge 4$, the lower and upper prefixes acquire the forced bits 0 and 1; their consecutiveness is preserved. Under the displayed child-gap conditions at the next row, this promoted stage therefore advances with coefficients determined by its retained prefixes.

Formalised by the two-coefficient gap recurrence and promote-and-advance theorem.

Proposition A.8 (quarter-band swallow obstruction)**.** *If a two-row update swallows a protected strip of bound B, and the lower coefficient pulse is at most the upper pulse plus 2, then*

$$B \le 4(g_+ - g_- + 1) + 2. \tag{A.1}$$

This is a necessary quarter-band condition, not a contradiction.

Formalised as the swallow bound.

Corollary A.9 (cofinal suffix-cylinder criterion)**.** *If full suffix-cylinder stages exist at arbitrarily large depths, then there is an infinite set $A \subseteq \mathbb{N}$ such that*

$$\sum_{n \in A} \frac{1}{2^n - 1} = \frac{1}{2}.$$

This is the cofinal-cylinder criterion.

Status and prior-art boundary. The two transition propositions do not supply their coefficient inequalities, and the finite base does not supply the cofinal family required by Corollary A.9. The strict-tail achievement-set literature supplies the surrounding Cantor-set geometry [14], not this finite carry recurrence; the comparison record makes no novelty claim for the recurrence itself.

A.5.4 Verified finite suffix-cylinder stages

The finite record has three milestones.

(1) *Depth* 27. Before the first half-divisor feedback row, a full selected window exists at every depth from 18 through 27, and the depth-27 endpoint covers the whole protected strip: stages through depth 27 and depth-27 strip coverage.

(2) *Depth* 29. The threshold-stage continuation still covers the strip and has a common suffix cylinder at every cutoff from 14 through 25, while cutoff 26 is explicitly not common: depth-29 strip coverage and cutoff-26 obstruction.

(3) *Depth* 52. An independent exact run reaches depth 51 with endpoint 51,327,745, beyond the next feedback head threshold, and the promoted successor at depth 52 is checked: depth-51 threshold margin and depth-52 terminal-strip witness.

These are finite milestones, not an unbounded induction; they do not provide the arbitrarily large stages required by Corollary A.9.

A.5.5 Half-divisor unit drop

At the doubled feedback row, changing only the terminal half-divisor bit from 0 to 1 increases the support coefficient by exactly one: the terminal-bit unit-drop identity. The same holds for any explicitly identified adjacent boundary pair, the boundary-pair unit-drop identity. The inverse-parent recurrence transfers such a pair to the next protected seam while retaining the exact unit-drop witness, the rewind unit-drop transfer and the protected-seam transfer. This is the local coefficient input for the rewind-seam profile, not a global construction of such boundary pairs.

A.5.6 A finite rewind-boundary obstruction

The scalar one-row rewind data alone still does not supply that missing boundary predicate. Lean gives a concrete depth-26 selected two-word window whose depth-27 explicit step has rewind history [1], is a width-two seam pair, and satisfies the doubled-row identity, but has no depth-13 half-divisor boundary-pair witness: the depth-27 boundary-pair counterexample. Thus the protected-window buffer in the compactness criterion is a genuine hypothesis, not a consequence of the scalar seam facts alone.

A.5.7 Explicit carry coordinates at the final non-right transition

For every adjacent cut at rank $s \geq 5$, the transition carry of the terminal-augmented below support equals the concrete middle coordinate:

$$4\,\mathrm{remainder} - \mathrm{belowPulse} - 4,$$

and identifies the terminal-augmented above support with the negative upper coordinate

$$-\big(4\,\mathrm{overshoot} + \mathrm{abovePulse} + 4\big).$$

The generic row-word identity is row-word transition identity, with the below and above specialisations at middle-coordinate identity and upper-coordinate identity. These identities make the remaining quantitative task exact: they do not prove exponential lower bounds, sign persistence, or a universal escape mechanism.

A.5.8 Strategy boundary: Campbell shift synchronisation

A proposed route combines a finite Mersenne shift from a putative last greedy skip with Campbell's quarter-exponent prime progression. The exact parameter requirements cannot be synchronised: if the combined progression modulus is $d \geq 2$, the phase factor is at most d, the fatal-window prime cap is at most four times that phase, and Campbell's applicability requires d^4 below the cap, then the inequalities are inconsistent. Formalised: the period-freeze incompatibility and the parameter incompatibility.

The associated shifted-zero condition is not an independent hypothesis. It is exactly equivalent both to infinitely many actual greedy skips and to $1/2 \in \mathcal{A}$: the shifted-zero/skip equivalence and the shifted-zero/half-membership equivalence. This closes one synchronisation strategy and one reformulation; it does not advance either endpoint beyond the already stated half-achievement problem.

Appendix consequence. This appendix supplies the local implications behind Theorem 4.14, Proposition 4.15, and Corollary 4.16. It does not establish the corollary's global hypothesis. The exact handoff is either a cofinal window, seam, or terminal approximation producing $1/2 \in \mathcal{A}$, or both an exclusion of every actual middle transition with $C_D = -2, -1$ and a proof of $\Theta_D < C_D$ at every remaining non-(-3) middle transition. The bounded checks and local no-go theorems provide neither route.

B Auxiliary rationality criteria from binary carry systems

This appendix supplies the general results assembled in Theorem 4.4 and Corollaries 4.5–4.8, and the detailed greedy geometry cited after Theorem 4.3. Its new working objects are a tempered integer tail orbit, its Boolean Möbius coordinate, the wrap–excess decomposition modulo the odd denominator, and exact rational death certificates for the achievement set. Appendices B.1–B.4 describe necessary or equivalent forms of a hypothetical rational support; Appendix B.5 describes the ambient achievement set and its one-sided finite nonmembership test. None of these objects by itself excludes a rational value.

The reading order is: a generic tail-orbit rationality normal form in Appendix B.1; its Boolean–Möbius specialisation in Appendix B.2; divisor-coverage constraints in Appendix B.3; denominator-wrap, reciprocal-mass, and unbounded-state constraints in Appendix B.4; and the exact greedy geometry with one-sided finite death certificates in Appendix B.5.

The dependency spine is

$$
\begin{aligned}
\text{a rational support value} &\iff \text{a tempered integral tail orbit} \quad (\text{B.1}),\\
\text{a specified support fraction } p/q &\iff \text{a Boolean–Möbius carry certificate} \quad (\text{B.2}),\\
\text{an infinite rational support} &\implies \text{sublogarithmic zero gaps} \quad (\text{B.3}),\\
\text{an infinite rational support} &\implies \text{unbounded-state and mass constraints} \quad (\text{B.4}).
\end{aligned}
$$

The last two lines list independent necessary conditions, not converses and not contradictions. Appendix B.5 is orthogonal to this rationality chain: finite greedy death proves nonmembership in the achievement set, whereas survival through any finite depth proves nothing.

The prior-art boundary is theorem-family-specific. The integral carry recurrence has a direct antecedent in Wang–Grau Ribas [15]; the strict-tail geometry is recorded by Kovač–Tao [14]; Möbius inversion, repetend algebra, and divisor averaging are classical (see, for example, Apostol [6]). We make no claim of mathematical priority for the converse/rigidity, certificate-normal-form, or coupled reciprocal-mass/collision statements collected here.

B.1 Exact normal form: tail-orbit rigidity

This subsection supplies the generic equivalence consumed in Appendix B.2. It converts rationality into an integer recurrence with a boundary condition; it does not itself exclude such an orbit or impose the Boolean support constraint.

For a coefficient sequence $\gamma : \mathbb{N} \to \mathbb{N}$ with $\gamma(n) \le n$, write

$$X_\gamma = \sum_{n\ge1} \frac{\gamma(n)}{2^n}, \qquad T_\gamma(N) = \sum_{j\ge1} \frac{\gamma(N+j)}{2^j},$$

the series and its scaled tail after N binary digits. Call an integer sequence u a *tempered orbit* for γ with multiplier $v \ge 1$ if

$$u(N+1) = 2\,u(N) - v\,\gamma(N+1) \text{ for all } N, \qquad \text{and} \qquad u(N)/2^N \to 0.$$

The recurrence is exact binary long division against the coefficient stream; the boundary condition forbids the homogeneous solution. Wang and Grau Ribas use the integral carry recurrence forced by rationality for the weighted binary special case $\gamma(n) = nd_n$ [15]. The theorem below extends that forward mechanism to arbitrary nonnegative $\gamma(n) \le n$; its explicit converse and subexponential-orbit rigidity are the local additions, with no priority claim attached. Wang–Grau Ribas's positive-density theorem and its Erdős #260 corollary are not used or formalised here.

Theorem B.1 (tail-orbit rigidity)**.** *Let $\gamma(n) \le n$ for all n. Every tempered orbit is the scaled tail, $u(N) = v\,T_\gamma(N)$ for all N; and X_γ is rational if and only if a tempered orbit exists for some multiplier $v \ge 1$.*

Formalised: the scaled-tail orbit identity (rigidity) and the rationality–orbit equivalence (the criterion), restated in Mathlib's irrationality vocabulary at the non-irrationality–orbit equivalence. The engine is one observation: a real orbit with $d(N+1) = 2\,d(N)$ and $d(N) = o(2^N)$ vanishes identically (zero-tail rigidity).

The tempered boundary is essential: the homogeneous parasite $N \mapsto 2^N$ can be added to any orbit without disturbing the recurrence, so positivity of an orbit certifies nothing by itself, and positivity is nowhere advertised as an equivalent criterion. The linear-growth hypothesis covers both constants introduced in Section 1: $\varphi(n) \le n$ for the #249 coefficients, and $f_A(n) \le \tau(n) \le n$ for the #257 support coefficients $f_A(n) = \#\{a \in A : a \mid n\}$. The results below instantiate the criterion on supports. Its separate totient anti-compression consequence belongs to the #249 rank route in Appendix C.4.

B.2 Exact #257 coordinates: Boolean–Möbius carry certificates

This subsection has one preliminary boundary result and then one exact equivalence chain. The boundary result rules out extracting the desired Boolean support directly from the signs in $L(\mu) = 1/2$; it is not a premise of the carry equivalence that follows.

Proposition B.2 (Möbius-sign support no-go)**.** *Let*

$$A_- = \{d \ge 2 : \mu(d) = -1\}, \qquad A_+ = \{d \ge 2 : \mu(d) = 1\}.$$

Then

$$\sum_{d\in A_-} \frac{1}{2^d - 1} = \frac{1}{2} + \sum_{d\in A_+} \frac{1}{2^d - 1} \ge \frac{1}{2} + \frac{1}{63} > \frac{1}{2}.$$

In particular, the signed identity $L(\mu) = 1/2$ cannot be converted into a Boolean support of value $1/2$ merely by selecting the negative Möbius indices.

Formalised: the negative-Möbius decomposition and the strict lower bound. The term $d = 6$ supplies the displayed $1/63$. This closes one sign-truncation route only; it does not exclude another infinite Boolean support with value $1/2$.

The exact coordinate chain. At base 2 the support series of Section 4 is itself a binary coefficient series, $\sum_{n\in A} 1/(2^n-1) = X_{f_A}$ (the support-series/binary-series identity). Boolean Möbius inversion first identifies the support from its divisor coefficients; tail-orbit rigidity then identifies rationality with a tempered integer orbit. Theorem B.4 is exactly the composition of those two equivalences. The divisor transform and Möbius inversion are classical Lambert-series tools [6, 7, 8]; the contribution here is their Lean-checked composition with the tempered-orbit criterion.

Proposition B.3 (Boolean Möbius inversion, both directions)**.** *On positive integers, $f_A = \mathbf{1}_A * \zeta$ and $\mu * f_A = \mathbf{1}_A$, where $\mathbf{1}_A$ is the indicator of A. Conversely, every integer-valued arithmetic function f whose Möbius transform is Boolean, $(\mu * f)(n) \in \{0,1\}$ for all $n \geq 1$, satisfies $f = f_B$ for the support $B = \{n \geq 1 : (\mu * f)(n) = 1\}$ selected by that transform.*

Formalised: the positive-support zeta identity, the Möbius inversion identity, and, with no search or finiteness hypothesis, the converse the Boolean reconstruction theorem.

On the carry side, Theorem B.1 applies at $\gamma = f_A$: the support series is rational exactly when a tempered carry orbit for f_A exists (the rationality–carry equivalence); a displayed fraction p/q corresponds to an orbit U with multiplier q and initial value $U(0) = p$ (the support-fraction carry equivalence); and exact division recovers the coefficient from the orbit, $\big(2U(N) - U(N+1)\big)/q = f_A(N+1)$ (carry-quotient recovery). The elementary divisor-pair bound $\tau(n) \leq 2\lfloor\sqrt{n}\rfloor$ confines the tails to a square-root strip, $T_{f_A}(N) \leq 2\sqrt{N} + 4$. For a nonempty support the orbit states are positive and satisfy $U(N) \leq q\,(2\sqrt{N} + 4)$ (the carry upper bound); and the strip is strong enough to re-derive the tempered boundary, so the analytic side condition can be traded for one inequality. The two coordinate changes compose.

Theorem B.4 (certificate-level equivalence)**.** *Fix an integer p and an integer $q \geq 1$. There is a nonempty support A of positive integers with $\sum_{n\in A} 1/(2^n - 1) = p/q$ if and only if there is an integer sequence U with $U(0) = p$ such that: every $U(N)$ is positive; $U(N) \leq q\,(2\sqrt{N} + 4)$; q divides $2U(N) - U(N+1)$ for every N; and the Möbius transform of the quotient sequence $\big(2U(N) - U(N+1)\big)/q$ is Boolean. The support is not guessed: it is reconstructed from the certificate by Proposition B.3.*

Formalised as the normalised support-fraction certificate equivalence.

Example B.5 (the support $\{2,3\}$)**.** The support $\{2,3\}$ has value $\frac{1}{3} + \frac{1}{7} = \frac{10}{21}$ (the two-point series value), its orbit is the pure period-six cycle $10, 20, 19, 17, 13, 26$ with multiplier 21 (the tempered two-point orbit), and the Möbius transform of the carry quotient recovers exactly $\{2,3\}$ (support recovery).

These are coordinates, not by themselves a strategy: nothing here excludes infinite Boolean carry paths, and no finite search is promoted to a completeness argument.

B.3 Necessary condition I: sublogarithmic divisor coverage

This theorem is a one-way consequence of a rational support value and is used in Corollary 4.6. It limits zero gaps in the divisor-coefficient stream; it neither reconstructs a support nor contradicts the existence of an infinite rational support.

The coefficient $f_A(n)$ counts the elements of A dividing n. Say that a zero window of length h starts after $c + N$ when $f_A(c + N + j + 1) = 0$ for every $0 \leq j < h$. Thus no integer in that interval is divisible by an element of the support.

Theorem B.6 (sublogarithmic zero windows)**.** *Let A contain a positive integer and suppose*

$$\sum_{a\in A} \frac{1}{2^a - 1} = \frac{p}{2^c v}, \qquad p \in \mathbb{Z}, \quad c \in \mathbb{N}, \quad v \geq 1.$$

For every $\varepsilon > 0$ there is a constant $B \geq 0$, depending only on ε, c, and v, such that every zero window after $c + N$ satisfies

$$h \leq \varepsilon \log_2(N+1) + B, \qquad \log_2 x := \frac{\log x}{\log 2}.$$

The bound is uniform in A, p, N, and h.

Formalised: sublogarithmic zero-window bound.

The proof first establishes, for each integer $k \geq 2$, the explicit divisor estimate

$$\tau(n)^k \leq \left(k^{2^k}\right)^k n.$$

It transports the resulting $n^{1/k}$ bound through the binary tail and then uses the exact carry recurrence across a zero window. Raising the resulting inequality to the kth power absorbs the occurrence of h on the right and leaves a coefficient $1/(k-1)$ in front of the binary logarithm. Choosing k after ε gives the theorem. This controls gaps in divisor coverage; it does not assert a subpower bound for the carry state itself.

B.4 Necessary conditions II: wraps, mass, and unbounded states

The dependency order is

$$\text{doubling residues} \longrightarrow \text{wrap count} \longrightarrow \text{reciprocal-mass bound} \longrightarrow \text{collision strengthening.}$$

The final unbounded-state theorem uses the same common-multiple collision mechanism without a summability hypothesis. The one-wrap classification identifies the extremal residue cycles; it is a side branch, not a premise of the general mass bound.

Suppose now the support series equals $p/(2^c v)$ with v odd and A containing a positive element. Let U be the unshifted tempered orbit of Appendix B.2 with multiplier $q = 2^c v$. The odd-denominator coordinate used here is

$$u(N) = \frac{U(c+N)}{2^c} = v\, T_{f_A}(c+N);$$

this is the state denoted u_N in Theorem 4.4. Rationality makes $u(N)$ a positive integer satisfying $u(N+1) + v\, f_A(c+N+1) = 2\, u(N)$. Moreover, $u(N)$ reduces modulo v to the doubling residue $\bar{p}_N = p\, 2^N \bmod v$ (shifted natural-state existence). Doubling a least residue either stays under v or wraps past it exactly once, so each step emits a wrap digit $k_N = \lfloor 2\bar{p}_N/v \rfloor \in \{0, 1\}$, and over any cycle length h with $2^h \equiv 1 \pmod{v}$ the repetend identity holds:

$$\sum_{N<h} \bar{p}_N \;=\; v \cdot w, \qquad w = \sum_{N<h} k_N,$$

with $w \geq 1$ when p is coprime to $v > 1$. The formal development records the displayed equality as the wrap-count identity and its non-vanishing as the positive-wrap theorem.

Proposition B.7 (one-wrap classification)**.** *Let $v > 1$ be odd, p coprime to v, and $h \geq 1$ with $2^h \equiv 1 \pmod{v}$. If the cycle of p wraps exactly once, then $v = 2^h - 1$ and the starting residue is a power of two: $\bar{p}_0 = 2^a$ for some $a < h$.*

The one-wrap cycles are thus exactly the Mersenne repetends $2^a/(2^h - 1)$. Formalised: one-wrap classification, from the generic closed-cycle form one-wrap cycle classification, and instantiated at the true order $h = \mathrm{ord}_v(2)$ (odd-order one-wrap classification). The classification is algebraic; a finite check of 446 coprime starting residues across twelve modulus/order rows is retained as kernel-checked validation, not as the proof (446-start finite validation).

The analytic bridge is Cesàro averaging. Write $\rho(A) = \sum_{a\in A} 1/a$ for the reciprocal mass. When the reciprocal family is summable, the mean of $f_A(1), \dots, f_A(N)$ converges to $\rho(A)$ — the density of the multiples of a, summed over the support — and the mean of the scaled tails has the same limit (divisor-mean limit, coefficient-tail mean limit). The divergent case is carried as an explicit disjunction, not through a default value assigned to a divergent formal sum. Each tail state splits exactly as $u(N) = v\, e_N + \bar{p}_N$ with integral excess $e_N = \lfloor u(N)/v \rfloor \geq 0$.

Proposition B.8 (order-wrap lower bound, with exact excess mean)**.** *Let $A \subseteq \mathbb{N}_{>0}$ be nonempty and suppose*

$$X_A = \frac{p}{2^c v}, \qquad p \in \mathbb{Z}, \quad c \in \mathbb{N}, \quad v > 1 \text{ odd.}$$

Let $h = \operatorname{ord}_v(2)$, let w be the number of wraps in the least-residue cycle $p2^N \bmod v$ for $0 \leq N < h$, and set

$$u(N) = vT_{f_A}(c + N), \qquad e_N = \left\lfloor \frac{u(N)}{v} \right\rfloor.$$

Then $\sum_{a\in A} 1/a$ diverges or $\rho(A) \geq w/h$; if moreover p is coprime to v, then $w \geq 1$, so $\rho(A) \geq 1/\operatorname{ord}_v(2)$. In the summable case the bound is the periodic part of an exact identity: the Cesàro mean of e_N converges, and

$$\rho(A) \;=\; \frac{w}{h} \;+\; \lim_{N\to\infty} \frac{1}{N} \sum_{n<N} e_n.$$

The divergent-or-bounded conclusion is formalised by the wrap-ratio mass alternative, its coprime specialisation by the odd-order mass alternative, and the automatic excess limit in the fraction-facing coordinates above by the shifted excess-mean identity. A rational value thus ties the odd part of its denominator to the sparsity of its support: a summable support with small $\rho(A)$ can only display denominators whose odd part has small multiplicative order of 2.

Corollary B.9 (collision strengthening at common multiples)**.** *In the summable case, let $F \subseteq A$ be finite and let L be a common multiple of F. Then*

$$\rho(A) \;\geq\; \frac{w}{h} \;+\; \frac{\lceil (|F| - 1)/2 \rceil}{L}.$$

In particular, an infinite support whose value is a dyadic rational $p/2^c$ has $\sum_{a\in A} 1/a$ divergent or $\rho(A) > 1$.

The mechanism is a collision: at every positive multiple of L at least $|F|$ support divisors land on the same index, so $f_A \geq |F|$ there, while the exact carry equation $f_A(n) + e_n = k_{n-1} + 2e_{n-1}$ with $k_{n-1} \leq 1$ lets one step absorb at most one unit — so the preceding excess must spike to at least $\lceil (|F| - 1)/2 \rceil$, and the spikes recur with density $1/L$. The three steps are formalised by the shifted common-multiple bound, shifted wrap–excess recurrence, and dyadic reciprocal-mass alternative.

Theorem B.10 (carry states are unbounded over infinite supports)**.** *Let A be infinite with value $p/(2^c v)$, $v \geq 1$. Then the positive integer carry state $u(N) = v\, T_{f_A}(c + N)$ is unbounded: for every B there is an N with $u(N) > B$.*

The proof picks $2B + 1$ positive support elements; at a common multiple L of all of them, the local inequality $1 + v\,|F| \leq 2\,u(L - c - 1)$ pushes the state past B. The conclusion is the fraction-facing unbounded-state theorem; its local engine is the support-cardinality state bound. So a rational value over an infinite support cannot run on a bounded carry alphabet. That closes finite-state readings of the carry system; it does not close the problem. Nothing in the constraints above prevents an infinite support from satisfying every one of these constraints. The repetend identities and divisor averages used here are classical. We make no claim of mathematical novelty for the coupled reciprocal-mass bounds, collision strengthening, or global unboundedness statement.

B.5 Independent geometry and one-sided nonmembership certificates

Finally, consider the set of all values a base-2 support series can take. For $n \geq 1$ put $w_n = 1/(2^n - 1)$, let $T(n)$ be the mass strictly after exponent n, and let

$$\mathcal{A} = \Big\{ \sum_{n \in A} w_n \ : \ A \subseteq \mathbb{N},\ 0 \notin A \Big\},$$

the *Mersenne achievement set*, the exponent 0 being normalised away as analytically invisible. Coding by supports of positive exponents is literally the formal support series (support-value identity). In this language the base-2 #257 statement reads: the rational members of $\mathcal{A}$ are exactly the finite subset sums. Nothing below decides that. Kovač and Tao verify, for every fixed integer $t \geq 2$, that

$$\sum_{\ell > n} \frac{1}{t^\ell - 1} < \frac{1}{t^n - 1},$$

and deduce distinct Lambert subsums and a Cantor set [14]. At $t = 2$ this is the relevant Lambert-series instance of Kakeya's classical strict-tail criterion for subsum sets [9]. The local contribution is the checked greedy criterion, quantitative enclosure, and finite nonmembership certificates over that known geometry; it does not attribute those refinements to the cited sources or settle #257.

Theorem B.11 (superincreasing geometry and greedy membership)**.** *For every $n \geq 1$ the weight dominates the whole tail after it, $T(n) < w_n$, with the two-scale gap $w_n - T(n) = \frac{2}{3}4^{-n} + O(8^{-n})$ and explicit valid O-constant 3. Consequently a real x belongs to $\mathcal{A}$ exactly when $x \geq 0$ and every greedy residual of x is at most the remaining tail; and normalised support coding is injective, so each member of $\mathcal{A}$ has a unique support, which the greedy recursion recovers bit by bit.*

Formalised: strict-tail inequality, gap big-O estimate (explicit bound explicit gap bound), greedy survival criterion, and normalised support-value injectivity, transferred to the kernel series at normalised support-series injectivity. The nonnegativity guard in the membership criterion is necessary: without it a negative target would survive every level while lying outside $\mathcal{A}$.

The binary coding also permits the global geometry to be checked directly.

Proposition B.12 (fat-Cantor geometry)**.** *The achievement set $\mathcal{A}$ is compact, perfect, totally disconnected, and nowhere dense. Its Lebesgue measure is exactly one.*

Formalised: measure-one theorem, perfectness, total disconnectedness, and nowhere density. Kovač–Tao supply the strict-tail Cantor-set context cited above. The formalisation records these exact properties without making a novelty or priority claim for them.

Proposition B.13 (exact rational runs and finite nonmembership certificates)**.** *The greedy recursion executed in exact rational arithmetic agrees step by step with the real recursion under casting. For a rational target x, write $r_n(x)$ for its exact rational residual after level n and, for a lookahead $\ell \geq 0$, put*

$$\widehat{T}_{n,\ell} = \sum_{k=0}^{\ell-1} w_{n+k+1} + 2w_{n+\ell+1} \in \mathbb{Q}.$$

The sum is empty when $\ell = 0$, and always $T(n) < \widehat{T}_{n,\ell}$. Hence the finite inequality $\widehat{T}_{n,\ell} \leq r_n(x)$ soundly proves $x \notin \mathcal{A}$; its level-one instance shows $3/4 \notin \mathcal{A}$. A rational number already known to lie in $\mathcal{A}$ has computable support bits.

Formalised: rational–real greedy agreement, rational tail-enclosure bound, finite death-certificate soundness, three-quarters nonmembership, and rational-member support-bit equivalence. The certificates are one-sided: survival through any finite depth proves nothing about membership, and no decidability of membership in $\mathcal{A}$ is asserted. Neither the global geometry nor the finite death certificates exclude rational values for either open series.

Appendix consequence. The transfer to the main text is Theorem 4.4 followed by the four independent Corollaries 4.5–4.8. Rationality yields the exact carry recurrence; an infinite rational support must separately satisfy the unbounded-state theorem, the sublogarithmic divisor-coverage bound, and the applicable odd- or dyadic-denominator reciprocal-mass constraint. Appendix B.5 also supplies the finite death certificates used in Theorem 4.3. What is not proved is that the support attached to either target violates one of these necessary conditions. That contradiction, not another rational normal form, is the missing premise.

C Technical reductions for the totient-certificate branch

This appendix expands the supporting mathematics behind Proposition 5.4, Theorems 5.5 and 5.6, and Theorem 5.7. It contains exact cone-flatness and target-interval reformulations, finite residue projections, denominator and bit-lifting formulae, bounded certificate families, and rigorous obstructions to several candidate strategies. Each statement retains its own hypotheses and status. None proves the unbounded certificate supply (5.1).

The exact spine is

$$\begin{aligned} R_{N+h} - R_N \notin \mathbb{Z} &\iff \exists L \geq 1,\ \mathsf{Sep}(h, N, L), \\ S \notin \mathbb{Q} &\iff \forall h \geq 1\ \forall N_0\ \exists N \geq N_0\ \exists L \geq 1,\ \mathsf{Sep}(h, N, L). \end{aligned}$$

Appendix C.1 gives equivalent coordinates for this fixed-to-cofinal spine. Appendix C.2 distinguishes one-scale mechanisms from the conditional cofinal implications that use them. Appendices C.3–C.4 expose possible inputs to an unbounded supply theorem and close several shortcuts. None of the latter three sections proves the missing cofinal supply in the second line.

C.1 Exact cofinal normal forms and cone consequences

This subsection contains two different kinds of statement. Cone flatness is a necessary consequence of rationality that a finite joint refuter may contradict at chosen vertices. The diagonal-pincer identity and its cofinal full-target-avoidance corollary are the exact fixed-scale and global normal forms of the open supply. Only the latter corollary is equivalent to the irrationality endpoint.

Cone flatness and completeness. On the cone $\{kM_t : k \geq 1\}$, rationality forces a single fractional constant across the whole cone.

Proposition C.1 (cone flatness)**.** *If S is rational, then for every sufficiently large t there is $\theta_t \in [0, 1)$ such that the fractional part of the tail R_{kM_t} equals θ_t for every $k \geq 1$.*

Formalised: cone flatness under rationality.

Proposition 5.4 states fixed-parameter certificate completeness, and Theorem 5.5 gives its global pointwise form. Their additional role here is cone-level: a joint finite refuter can interrogate several cone vertices at once and is sharper than any pairwise test. It refutes the single-fractional-part model that rationality predicts on a finite sample (finite cone-nonflatness refuter). This finite cone predicate is sound in the direction just stated: a certificate produces a non-integral pair and hence contradicts the sampled flatness model. No converse is proved saying that every non-flat finite sample, or every irrational value of S, must be detected by this joint predicate. It is therefore a refuter, not a complete decision procedure.

An exact target interval. The diagonal calculation also admits a more local normal form. Put $H_t = M_t$ and

$$D_t = R_{2H_t} - R_{H_t}.$$

Separate an exact rational Möbius shadow $Q_t = a_t/d_t$ in lowest terms and define the remaining *foreign defect* by $F_t = D_t - Q_t$. The full target is the residue class of $-a_t$ modulo d_t:

$$\mathsf{Hit}(t) \quad \Longleftrightarrow \quad \exists k \in \mathbb{Z}, \qquad d_t F_t = -a_t + d_t k.$$

Since $D_t = a_t/d_t + F_t$, this is exactly the condition $D_t \in \mathbb{Z}$; no complementary term is discarded or assigned a conjectural sign.

Proposition C.2 (diagonal pincer decomposition)**.** *For every t, diagonal integrality is equivalent to the foreign diagonal defect meeting the full target.*

This is the full-target equivalence.

Corollary C.3 (exact full-target avoidance characterisation)**.** *The following are equivalent:*

$$\begin{aligned}&S \notin \mathbb{Q};\\ &\forall\, t_0 \in \mathbb{N}\ \exists\, t \geq t_0, \qquad \neg\mathsf{Hit}(t).\end{aligned}$$

The reverse implication is the unbounded target-avoidance criterion. For the forward implication, Theorem 5.7 supplies cofinally many non-integral diagonal differences, and Proposition C.2 converts each one into a missed full target. Thus the proposition is the fixed-scale equivalence and the corollary is its exact global form. The cofinal avoidance statement remains open.

C.2 Fixed-scale mechanisms and conditional cofinal implications

This subsection is organised by quantifier. Its enclosures, predicates, checked certificate families, and projection equivalences are finite or one-scale mechanisms. Whenever a corollary concludes irrationality, its additional premise is an unproved cofinal supply of the relevant finite mechanism. No result here constructs such a supply or fills the missing quantifier in Appendix C.1.

A one-way analytic enclosure. Truncate the Möbius-square identity of Proposition D.6 after D terms. The resulting rational quantity gives a Lambert-projected centre $C_{H,D}$ for the scaled diagonal difference. The error is exactly a signed squared-Mersenne tail:

$$\mathsf{scaledDiagonal}(H) - C_{H,D} = c_H \sum_{d>D} \frac{\mu(d)}{(2^d-1)^2},$$

where c_H is the explicit diagonal coefficient in the formal source. The absolute value of the sum is bounded geometrically by $4/[3(2^{D+1}-1)^2]$. Thus a rational separation of the centre from every integer by more than the scaled bound proves target avoidance; the argument does not estimate the sign of the tail.

Proposition C.4 (squared-Mersenne enclosure)**.** *The scaled diagonal tail difference differs from its Lambert-projected centre by the squared-Mersenne tail. Separation of the centre from the full target by more than the proved tail bound forces target avoidance.*

Formalised: squared-Mersenne tail identity and Lambert-projected separation criterion. The enclosure is one-way: separation at a chosen truncation forces target avoidance, but target avoidance is not asserted to force that separation.

A reduced-denominator finite predicate. Separately, there is a stricter finite condition for an odd reduced denominator u. Its predicate $\mathsf{UnitGap}(u, N, K)$ asks that every candidate integer in the associated dyadic tail interval be non-coprime to u. At a prime power $u = p^e$, the condition is equivalent to the exact alternative that the interval has no candidate, or that it has one candidate and p divides that candidate. Thus such an interval has at most one candidate.

Proposition C.5 (prime-power unit-gap ceiling)**.** *For every prime power p^e with $e > 0$, a* $\mathsf{UnitGap}(p^e, N, K)$ *has candidate count at most one. The row $(u, N, K) = (3, 3, 5)$ satisfies this predicate, although its corresponding ordinary empty-gap inequality fails.*

Formalised: prime-power unit-gap characterisation, prime-power one-candidate bound, three-window unit-gap certificate, and three-window ordinary-gap failure. This is a finite certificate-level strengthening. We do not establish that rationality of S supplies these predicates at unbounded parameters.

The certificates are non-empty. The initial segment of the required supply is machine-checked, so the reduction is not vacuous. By completeness, each of these is a kernel-checked proof of a specific non-integral tail difference, not a numerical experiment.

Example C.6 (verified finite instances, expanded record)**.** The following are checked by the Lean kernel:

- $\mathsf{Sep}(h, 12, 16)$ for every period $h \in [1, 8]$;
- finite certificates for every period $h \in [1, 16]$ at a fixed window;
- diagonal certificates $\mathsf{Sep}(M_t, M_t, L)$ at $t = 1$ and the 27 strict lcm-jump scales through 64, namely $t \in \{1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32, 37, 41, 43, 47, 49, 53, 59, 61, 64\}$;
- a tabulated joint cone-vertex certificate.

The four validation classes are formalised, in order, by the small-window certificate family, periods through sixteen, 28 imported diagonal certificates through scale 64, and cone-cell certificate table. These instances show that each layer of the reduction is non-vacuous and, by completeness, exact. They are not evidence that the unbounded supply exists.

The first row of this table is worked in full following Example 5.8. The soundness theorem (certificate soundness) converts each successful row into one exact non-integral tail difference; no row asserts that such differences occur at unbounded scales.

Finite foreign-residue projection. There is a second finite presentation of the same obstruction. For a cutoff D, split the finite Möbius residue diagonal into the channels $d \leq D$ which divide H and the remaining channels. The divisor part is exactly the truncated Möbius shadow, while every retained nondivisor channel is explicit. Above the stable cutoff $2H$, a finite window of omitted channels is bounded by

$$c_H \left(\frac{2}{2^D} + \frac{4}{3\, 4^D} \right),$$

where c_H is the diagonal coefficient. Thus, if the actual foreign defect agrees with this projection within the displayed bound, and the projected quantity is farther than that bound from every integer, then the full target is missed.

Proposition C.7 (finite foreign-residue projection)**.** *Let $F_H^{(D)}$ be the retained nondivisor projection at cutoff D, let F_H be the actual foreign defect, and let Q_H be the scaled explicit divisor shadow. The finite residue diagonal is exactly the sum of $F_H^{(D)}$ and its divisor-channel complement. For $2H \leq D$, put*

$$B_{H,D} = c_H \left(\frac{2}{2^D} + \frac{4}{3\, 4^D} \right).$$

If

$$|F_H - F_H^{(D)}| \leq B_{H,D} \quad \text{and} \quad |Q_H + F_H^{(D)} - z| > B_{H,D} \quad \text{for every } z \in \mathbb{Z},$$

then the full target at height H is not attained.

Formalised: foreign–divisor channel split, omitted-channel bound, and projected-separation criterion. The passage from finite windows to the analytic foreign complement, including the corresponding kernel identity, is not supplied here. Nor is an unbounded family of separated projections; these are parts of the open certificate-supply problem.

Second, the rational Möbius shadow has an exact reduced denominator. If r_t is the square-free kernel of H_t, $s_t = H_t/r_t$, and $J(r_t)$ is the odd Jordan scalar defined in the algebraic shadow chain, then

$$d_t = \frac{2^{r_t} - 1}{\gcd\bigl(2^{r_t} - 1,\, s_t|J(r_t)|\bigr)}.$$

For $t \geq 5$, the product of the Mersenne factors $2^p - 1$ over primes $t/2 < p \leq t$ divides d_t; in particular $d_t \geq 2^{\lfloor t/2 \rfloor}$. This is a denominator theorem for Q_t only. Cancellation by F_t is exactly what the full-target condition retains.

Proposition C.8 (Mersenne-shadow denominator)**.** *Let r_t be the square-free kernel of H_t, let $s_t = H_t/r_t$, and let $J(r_t)$ be the odd Jordan scalar. The reduced denominator d_t of the scaled Möbius shadow is*

$$d_t = \frac{2^{r_t} - 1}{\gcd\bigl(2^{r_t} - 1,\, s_t|J(r_t)|\bigr)}.$$

If $t \geq 5$, then

$$\prod_{\substack{p \text{ prime} \\ t/2 < p \leq t}} (2^p - 1) \;\mid\; d_t, \qquad d_t \geq 2^{\lfloor t/2 \rfloor}.$$

Formalised: upper-half Mersenne divisor bound and exact shadow denominator.

Finally, the remaining diagonal error can be represented by a finite integer projection and then by binary suffix data. At depth L, let

$$P_L(M) = \sum_{j=0}^{L-1} \varphi(M + 1 + j)\, 2^{L-1-j}$$

be the length-L window numerator underlying Definition 5.3, and define

$$W_{t,L} = P_L(2H_t) - P_L(H_t), \qquad \eta_{t,L} = W_{t,L} \bmod 2^L,$$

with the residue chosen in $[0, 2^L)$. The finite residue certificate is the asymmetric safe interval

$$H_t + L + 2 < \eta_{t,L} < 2^L - (2H_t + L + 2).$$

The two unequal margins are the tail bounds at the two cone vertices H_t and $2H_t$; replacing them by a symmetric slogan would lose part of the checked statement.

Corollary C.9 (finite residue projection)**.** *For every $t, L \in \mathbb{N}$, the diagonal projection certificate is equivalent to*

$$H_t + L + 2 < \eta_{t,L} < 2^L - (2H_t + L + 2),$$

and either condition forces full-target avoidance at height t. Consequently,

$$\forall t_0 \in \mathbb{N}\; \exists t \geq t_0\; \exists L \in \mathbb{N}, \qquad H_t + L + 2 < \eta_{t,L} < 2^L - (2H_t + L + 2)$$

implies that S is irrational.

Formalised: residue–projection equivalence and unbounded projection criterion. The finite equivalence is only between the residue inequality and the projection certificate. Neither condition is asserted to be necessary for full-target avoidance or for irrationality. The stronger adjacent-suffix supplies formalised in the same module imply the displayed cofinal supply; they are not additional conclusions of one finite certificate. Finite instances of this implication are proved, but no unbounded projection or suffix supply is established.

Primality certificates at changing-lcm scales through $t = 64$. A finite family of Lucas certificates proves the primality facts required by the diagonal-pincer certificates at every changing-lcm scale through $t = 64$; for example, representative Lucas primality certificate. This is a finite theorem family, not an unbounded certificate supply or an irrationality proof.

The diagonal certificate at $t = 64$. Using those primality facts, Lean checks the individual diagonal-pincer certificate at $t = 64$, scale-64 diagonal certificate. This is one finite endpoint certificate, not an unbounded supply.

C.3 Candidate cofinal inputs and no-go boundaries

The three-rank condition below is a candidate cofinal sufficient condition, while its affine form exposes a precise obstruction to obtaining that condition from uniform correction bounds alone. The remaining results are independent filters, not consecutive proof steps: the fresh-prime decomposition and one-bit lift expose exact correction terms that an unbounded argument would have to control, and the parity, prime-adjunction, scalar-localisation, and square-CRT results close four natural shortcuts. No no-go result consumes the preceding filter as a hypothesis.

A three-rank odd window. At an analytically admissible odd power-of-two rank, centrality at any one of the canonical odd rank or its next two base-four successors yields the required adjacent-suffix gap condition, three-rank suffix-gap implication. Thus a cofinal three-rank band supply implies irrationality, cofinal three-rank criterion; that supply is not proved.

Affine form of the three-rank condition. With the stated short-correction bounds, the three central-band predicates are equivalent to one three-scale affine escape condition, three-scale affine equivalence. A zero-start affine orbit shows that a uniform correction envelope alone does not force escape at any finite number of ranks, zero-orbit obstruction.

Fresh-prime deficit decomposition. The literal foreign channel also has an endpoint form suited to curvature tests. Retain only the prime support already present at H in the two endpoints $H + s$ and $2H + s$, and write the resulting discrepancy from the actual totient as an endpoint deficit. Each such deficit is nonnegative. The actual diagonal increment then equals the old-prime increment plus the lower deficit minus the upper deficit. The second and five-point curvature operators preserve this exact split; the latter has coefficient row $[-8, 4, 2, 1, 1]$. Consequently, the old curvature margin loses at most the stated adverse weighted deficit.

Proposition C.10 (fresh-prime deficit decomposition)**.** *The endpoint fresh deficits are nonnegative, give an exact decomposition of the foreign channel, and bound the loss from the old-prime curvature margins.*

Formalised: endpoint-deficit nonnegativity, foreign-increment decomposition, and adverse-deficit curvature bound. This supplies the exact correction term for the finite evaluations. It supplies neither a cofinal deficit bound nor an irrationality criterion.

One-bit lifting. The power-of-two signed-margin condition admits a separate exact arithmetic reduction. If a signed difference is already divisible by 2^k, lifting its congruence to 2^{k+1} is equivalent to evenness of the quotient after division by 2^k. Failure of the lift gives the unique translated residue class. In particular, a displayed difference $P - N = 16c$ agrees modulo 32 exactly when c is even; otherwise the residue is the offset class by 16.

Proposition C.11 (one-bit signed-margin lift)**.** *The fifth-bit question after fourth-bit agreement is exactly one cofactor parity test, with the complementary offset class characterising failure.*

Formalised: one-bit divisibility lift, mod-32 cofactor criterion, and offset-16 complement. The centred-state formulation identifies the signed-margin interval condition with failure of this next lift, subject to its stated correction and margin budget: centred-lift equivalence and centred mod-32 equivalence. This is a finite arithmetic equivalence. It does not supply the required cofactors at unbounded scales and gives no irrationality conclusion.

Parity is not a margin certificate. The preceding lift does not settle the reduced signed-margin interval condition. For a positive even modulus m and integral radius parameter a with $16 < a < 8m - 16$, each parity class contains both a residue inside the interval and one outside it. The same statement applies to a positive power-of-two modulus. Thus even a fifth-bit computation, by itself, cannot decide the required margin condition.

Proposition C.12 (parity does not decide the reduced margin condition)**.** *Let $m, a \in \mathbb{Z}$, with $m > 0$ even and*

$$16 < a < 8m - 16,$$

and put

$$\mathcal{M}_{m,a} = \{y \in \mathbb{Z} : \exists k \in \mathbb{Z},\ 16|y - km| < a\}.$$

Each parity class contains both a member and a nonmember of $\mathcal{M}_{m,a}$. In particular, the conclusion holds for $m = 2^j$ whenever $j \geq 1$.

Formalised: parity obstruction and power-of-two parity obstruction. The result is structural and proves no signed-margin certificate at any particular scale; it excludes parity as a sufficient standalone criterion.

Prime-adjunction no-go theorem. One tempting way to seek an unbounded obstruction is to compare the full target at H, pH, qH, and pqH. The following affine transport law rules out this route before any projection is introduced. For every positive k, the actual diagonal satisfies an affine transport law

$$D_{kH} = Q_k(H)D_H + Z_k(H), \qquad Q_k(H) \in \mathbb{N}, \quad Z_k(H) \in \mathbb{Z}.$$

Hence integrality at H propagates to every multiple, and the four target hits at the prime-adjunction diamond are equivalent to the single hit at its root (prime-adjunction diamond collapse). The corresponding transport of the complete foreign defect has zero diamond curvature (foreign-correction flatness). Thus explicit-shadow fibres alone cannot create a holonomy obstruction. A future positive route would need a separately defined projection together with a theorem controlling what that projection discards. This is a proved obstruction to one strategy, not progress on the unbounded-supply proposition.

Proposition C.13 (scalar-localisation obstruction)**.** *Let $x \in \mathbb{Q}$ and $c \in \mathbb{Z}$. If $H \mid \operatorname{den}(x)$ and the reduced denominator of cx divides H, then*

$$\frac{\operatorname{den}(x)}{H} \mid |c|.$$

Formalised: complementary-denominator divisibility and scalar-localisation identity. Thus a scalar localisation can move a complementary denominator factor into its coefficient, but cannot erase it. This rules out a scalar-denominator shortcut of that form. It neither proves full-target avoidance nor supplies any certificates, so it gives no irrationality criterion for S.

Proposition C.14 (square-CRT correction suppression)**.** *Let E be a finite family indexed by distinct primes p_i, with prescribed anchors A_i and residues a_i. There is a bounded common base n such that, for each $i \in E$ and each shift h with $p_i \nmid a_i + h$, one has*

$$n = A_i + p_i(a_i + p_i t) \quad \textit{and} \quad \varphi(n + p_i h - A_i) = (p_i - 1)\varphi(a_i + p_i t + h)$$

for a suitable $t \in \mathbb{N}$. In particular, if every p_i exceeds a fixed horizon J, the choice $a_i = 0$ works simultaneously for all $1 \le h \le J$.

Formalised: finite clean totient family and clean horizon family. The finite mechanism does not force a cube coefficient to be nonzero: there is a checked clean two-step cube with both displayed coefficients zero (vanishing clean block) and a separate checked clean cube with second coefficient -4 (nonzero clean block). Thus the result gives neither an unbounded certificate supply nor an irrationality criterion for S.

C.4 Unconditional dyadic rank and carry anti-compression

The determinant producer in this route is unconditional. At every level, a CRT construction prescribes one target prime channel and fresh off-target prime divisors; Dirichlet's theorem supplies the required primes in the resulting progressions. After the exact powers of two are divided from the columns, the evaluation matrix is the identity modulo two. This proves a nonzero separated minor at every level rather than merely postulating one.

Theorem C.15 (infinite dyadic totient-kernel rank)**.** *For every $e \ge 0$, the canonical level-e dyadic totient-kernel family is linearly independent over $\mathbb{Q}$ and its span has dimension $2^e + 1$. Consequently, the span of all sections*

$$n \longmapsto \varphi(2^j n + r), \qquad j \ge 0, \quad 0 \le r < 2^j,$$

is not finite-dimensional over $\mathbb{Q}$.

Formalised by the all-level minor producer CRT–Dirichlet separated minors, the independence theorem canonical kernel independence, the exact rank formula exact canonical rank, and infinite full-kernel dimension. The theorem concerns the coefficient kernel. It does not imply that S is irrational.

Proposition C.16 (finite-level carry anti-compression)**.** *If the totient coefficient series were rational, there would be a positive- multiplier tempered integral carry orbit whose canonical carry-section span has dimension at least $2^e - 1$ for every e.*

Formalised: the carry-kernel rank lower bound and the unconditional unbounded-rank consequence of rationality. No theorem bounds the dyadic section rank of every rationality-supplied tempered carry. The proposition therefore identifies an infinite rationality-side barrier, not an irrationality criterion for S.

Proposition C.17 (rectangular determinant expansion)**.** *For matrices $M : \iota \times \kappa \to R$ and $N : \kappa \times \iota \to R$ over a commutative ring, the determinant of MN is the sum over all maps $p : \iota \to \kappa$ of the determinant of the corresponding selected-column matrix times the diagonal product $\prod_i N_{p(i),i}$.*

This is the rectangular determinant expansion.

Proposition C.18 (dominant dyadic term)**.** *If one selected integer numerator is odd and its dyadic exponent is strictly larger than every other selected exponent, then the common numerator after dyadic denominator clearing is odd and hence nonzero.*

Formalised by dominant dyadic parity, and dominant dyadic nonvanishing. These are finite algebraic inputs to a signed-moment route. They do not show that any actual totient Hankel determinant is nonzero, and they give no irrationality criterion for S.

The formal proof also exposes the producer-neutral separated-minor rank criterion. The two generic algebraic propositions above remain useful as possible inputs to other determinant routes, but the totient-kernel minor itself is supplied at every level by Theorem C.15.

Proposition C.19 (even-residue reduction)**.** *Every even residue channel in the canonical dyadic totient-kernel family is exactly a scalar multiple of its odd-core channel.*

Formalised as the even-residue reduction. This removes duplicate channels before the separated minor is constructed.

Proposition C.20 (compressed-adjoint contradiction)**.** *A compressed-adjoint certificate whose nonzero multiple is strictly smaller in absolute value than its positive modulus is inconsistent.*

Formalised as the compressed-adjoint contradiction. This is a conditional finite contradiction, not a source of certificates at every level.

Proposition C.21 (residual-gauge obstruction for monomial minors)**.** *Let a monomial matrix have a common nonzero residual weight in each column. The residual weights act by a right diagonal gauge, and therefore preserve determinant nonvanishing. If one exponent is* 1*, the inverse-phase locked gauge makes that row identically* 1 *while preserving a nonzero determinant whenever the coefficient-side determinant is nonzero. If the residual is allowed to depend on both row and column, it reconstructs a consecutive-power matrix whose zeroth row is identically* 1*.*

Formalised: residual-gauge determinant equivalence, locked-reconstruction nonvanishing, and reconstructed zero row. Thus a residual-blind rank, determinant, or conditioning criterion alone cannot exclude the target. The proposition does not exclude a criterion that also uses an arithmetic coupling identity, and it gives no irrationality criterion for S.

Appendix consequence. Nothing in this appendix strengthens the fixed-parameter completeness of Proposition 5.4 into an unbounded supply. Its exact handoff is therefore the still-open quantifier in Theorem 5.6:

$$\forall h \geq 1 \; \forall N_0 \; \exists N \geq N_0 \; \exists L \geq 1, \qquad \mathsf{Sep}(h, N, L).$$

The one-scale mechanisms and conditional implications in Appendix C.2, the candidate inputs and no-go boundaries in Appendix C.3, and the conditional rank routes in Appendix C.4 delimit possible approaches to this statement. None provides the required cofinal certificates.

D Lambert, probability, and finite-algebraic complements

This appendix supplies the extensions cited from the Mersenne–Lambert ladder in Section 3 and from the coprime-pair representation after Proposition 5.2. Its new results are the periodic-weight endpoints, finite denominator obstructions, the Möbius-square and squared Lambert identities, and the Stern–Brocot cylinder and continuant laws. These enlarge the two coordinate pictures; they are not additional steps in either open proof spine.

The three families are parallel, not cumulative. Appendix D.1 proves or classifies weighted Lambert series with coefficient inputs different from the open totient input. Appendix D.2 gives exact identities and finite algebraic boundaries for the squared and Möbius coordinates. Appendix D.3 develops unconditional geometry internal to the coprime-pair probability coordinate. No conclusion about a comparison input transfers irrationality to S, and none of the three families supplies the cofinal Sep supply theorem.

D.1 Weighted Lambert comparison families

This subsection changes the coefficient input. Its eventually-periodic theorems settle or classify those comparison series, while the period-four example blocks one residue-blind shortcut. None has the totient weight defining S, so no irrationality status transfers to Erdős #249.

Periodic weights. For $L_b(\gamma) = \sum_{a \geq 1} \gamma(a)/(b^a - 1)$, the checked endpoints are:

Theorem D.1 (eventually-periodic nonnegative weights)**.** *An eventually-periodic nonnegative rational weight with a positive periodic value gives an irrational series.*

Formalised: eventually-periodic nonnegative criterion. This is contained in a broader theorem of Luca–Tachiya [13].

Theorem D.2 (periodic signed-weight dichotomy)**.** *For a periodic integer weight, the series is irrational or terminating in base b.*

Formalised: periodic signed-weight dichotomy.

Proposition D.3 (residue-blind obstruction)**.** *For the period-four weight* $1, 0, -1, 0$*, the divisor coefficient vanishes on* $n \equiv 3 \pmod 4$*.*

Formalised: period-four coefficient obstruction.

D.2 Finite algebraic boundaries and squared-Lambert coordinates

This subsection keeps the relevant coordinates but separates three roles: finite denominator-clearing obstructions, the exact Möbius-square identity for S, and comparison rows in the squared-Lambert calculus. The identity changes coordinates; the finite propositions delimit algebraic methods; and the status of one squared-Lambert row does not pass to another.

Finite algebraic boundaries. Put $a_n = (\varphi * \mu)(n)/n$. If an integer D clears these coordinates through N, the exact obstruction is:

Proposition D.4 (primitive-coordinate finite-jet obstruction)**.** *D is divisible by an explicit two-tier primorial T_N; hence no fixed D clears every a_n.*

Formalised: two-tier primorial obstruction.

For $d \geq 1$, define the pointed periodic atom

$$\omega_d(N) = \frac{2^{N \bmod d}}{2^d - 1}.$$

It satisfies the exact carry law

$$2\omega_d(N) - \omega_d(N+1) = \mathbf{1}_{d \mid N+1}.$$

Proposition D.5 (finite Mersenne-atom determinant boundary)**.** *Fix $m \geq 1$, conductors $d_j \geq 1$, indices $n_{ij} \geq 0$, and integer column weights w_j, and set*

$$W_{ij} = w_j \omega_{d_j}(n_{ij}), \qquad U_{ij} = 2^{n_{ij} \bmod d_j}.$$

Then

$$\left(\prod_{j=1}^{m} (2^{d_j} - 1) \right) \det W = \left(\prod_{j=1}^{m} w_j \right) \det U \in \mathbb{Z}.$$

If the integer on the right is nonzero, then the absolute value of the cleared determinant on the left is at least 1*.*

Formalised: Mersenne-atom carry law, determinant-clearing identity, and cleared-determinant lower bound. Thus finite Mersenne-atom determinants recover exactly their explicit column denominators; the clearing creates no additional height surplus. These are scoped finite-denominator obstructions, not irrationality criteria for S.

Proposition D.6 (the Möbius-square lens)**.**

$$S \;=\; \tfrac{1}{2} \;+\; \sum_{d\geq 1} \frac{\mu(d)}{(2^d-1)^2}.$$

Formalised: Möbius-square identity. Writing $T = \sum_{d\geq 1} \mu(d)/(2^d-1)^2$, irrationality of S is equivalent to irrationality of T, since $S = \frac{1}{2} + T$. The factor $1/(2^d-1)^2$ is the probability that d divides two independent fair-coin waiting times (Section 5.2). The signed constant, the coprime-pair mass, and the tail differences of Section 5.3 are three readings of the same quantity.

Squared Lambert calculus. For $L_2(f) = \sum_{d\geq 1} f(d)/(2^d-1)^2$, the denominator is the probability that d divides two independent geometric waiting times. The formal transfer theorem, squared Lambert transfer, therefore gives a divisor calculus for their gcd.

Proposition D.7 (squared Lambert ladder)**.** *At $r = 1/2$, the three rows $f = \mu, \mathbf{1}, \varphi$ yield respectively*

$$L_2(\mu) = S - \tfrac{1}{2}, \qquad L_2(\mathbf{1}) = \sum_{n\geq 1} \frac{\sigma(n)-\tau(n)}{2^n}, \qquad L_2(\varphi) = \sum_{n\geq 1} \frac{P(n)-n}{2^n},$$

where P is Pillai's gcd-sum function.

Formalised: totient gcd-moment identity. Postelmans–Van Assche imply irrationality of the middle row [12]; this does not transfer to the Möbius row, which is exactly the open constant $S - \frac{1}{2}$.

D.3 Probability-coordinate geometry: reduced slopes and cylinders

These results live entirely inside the coprime-pair probability coordinate. They normalise reduced directions and quantify Stern–Brocot cylinder and run geometry. They are unconditional geometric consequences and do not imply the open certificate supply.

Reduced directions and cylinder masses. The fair-coin pair has a particularly rigid distribution after division by its gcd. For positive coprime a, b, put

$$\nu(a,b) = \frac{1}{2^{a+b}-1}, \qquad M(a,b) = \frac{1}{(2^a-1)(2^b-1)}.$$

The first quantity is the probability that the reduced direction is (a,b); the second is the mass of the Stern–Brocot cylinder rooted at (a,b). Let $M_d(a,b)$ be the total stop mass in the first d generations below that root, where a node (u,v) has stop mass $(2^{u+v}-1)^{-1}$ and children $(u+v,v)$, $(u,u+v)$.

Proposition D.8 (reduced-direction normalisation)**.** *The reduced directions form a probability distribution,*

$$\sum_{\substack{a,b\geq 1\\(a,b)=1}} \nu(a,b) = 1.$$

This is the reduced-direction mass one.

Theorem D.9 (Stern–Brocot cylinder recursion and convergence)**.** *At every positive node,*

$$M(a,b) = \frac{1}{2^{a+b}-1} + M(a+b,b) + M(a,a+b),$$

and, uniformly in a, b and d,

$$0 \le M(a,b) - M_d(a,b) \le \left(\frac{2}{3}\right)^d M(a,b).$$

Consequently $M_d(a,b) \to M(a,b)$; at the root $M(1,1) = 1$.

Formalised: cylinder mass recursion, depth-d remainder bound, and cylinder convergence. The recursion is the elementary rational identity obtained after setting $A = 2^a$, $B = 2^b$; the quantitative point is that the two children retain at most two thirds of their parent's mass.

Alternating-run stability. Write an alternating Stern–Brocot word as r nonempty runs of lengths $n_i = 1 + e_i$, with $e_i \ge 0$. Starting with $P_\varnothing = (1,1)$, define

$$P_{(n_1,\dots,n_r)} = (n_1 A + B, A) \quad \text{when} \quad P_{(n_2,\dots,n_r)} = (A,B), \qquad H(n_1,\dots,n_r) = P_1 + P_2.$$

Thus H is the arithmetic height of the primitive pair reached by the word. Let $F_0 = 0, F_1 = 1$.

Theorem D.10 (Fibonacci–continuant height stability)**.** *Every r-run word satisfies*

$$H(1+e_1,\dots,1+e_r) \ge F_{r+3} + F_{r+1} \sum_{i=1}^{r} e_i,$$

and $H(1,\dots,1) = F_{r+3}$. A defect confined to position $i \in \{0,\dots,r-1\}$ has the exact size

$$H(\underbrace{1,\dots,1}_{i}, 1+e, \underbrace{1,\dots,1}_{r-i-1}) = F_{r+3} + F_{i+2} F_{r-i+1} e.$$

More generally, the checked recurrence gives the full positive multiaffine continuant defect.

Formalised: Fibonacci height lower bound, multiaffine run-height formula, aggregate defect lower bound, one-site defect identity.

Proposition D.11 (one-run probability mass)**.** *The two orientations with one nonempty Stern–Brocot run have total mass*

$$\frac{1}{2}.$$

This is the one-run mass.

Proposition D.12 (unit-spine denominator exponent)**.** *The natural denominator exponent on the all-unit r-run spine is*

$$\sum_{j=0}^{r-1} F_{j+2} = F_{r+3} - 2.$$

This is the natural denominator exponent.

The Lambert and Möbius identities used above are classical [6, 7, 8], and the mediant language goes back to Farey [10]. Postelmans–Van Assche supply the cited irrationality of the constant-weight squared-Lambert row, not these cylinder or run statements. We claim the displayed Lean-checked identities and estimates, not priority for their assembled package. In particular, the exact equality of the Fibonacci height and denominator scales is a boundary, not an irrationality argument: no theorem here proves that a run tail survives denominator clearing.

Appendix consequence. The transfer is coordinate extension only. Appendix D.1 enlarges the Mersenne–Lambert comparison family in Section 3; Appendix D.2 supplies the Möbius-square and gcd-moment coordinates; and Appendix D.3 refines the coprime-pair representation in Proposition 5.2. These are three independent handoffs. None implies the cofinal Sep supply for S or a proof of $1/2 \in \mathcal{A}$. Those, rather than any result internal to this appendix, remain the missing main-spine inputs.

E Guide to the formal sources

The paper is meant to be read as mathematics. The blue mathematical phrases following formalised statements are intended as links to exact checked theorems in the final published source revision. In this draft the common source pin is a release blocker. Exact declaration names remain in the machine-readable audit layer.

For a first reading, the principal landmarks are Theorem 5.1 for the totient denominator exclusion; Proposition 5.4 and Theorems 5.5–5.7 for the exact certificate normal forms; and Theorems 4.11 and 4.14 for the two half-value classifications. Theorem 4.1 is the classical closed comparison case. For the strongest independent structure beyond the two open spines, see Theorem C.15, the sublogarithmic zero-window corollary in the main #257 results, and Theorems D.9 and D.10. The public repository contains a complete searchable cross-reference for readers who need declaration-level detail.

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